

Concept 1 | Coordinate Geometry Basics

English Student Edition • Focused formulas and guided practice

Formula opener

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Classified Practice

Coordinate Geometry Basics

Q001 **Written**

A (−4), B (8): **distance AB**

Q002 Written**A (−5), B (7): distance AB**

Classified Practice

Coordinate Geometry Basics

Q003 **Written**

For two points A (-2) and B (6) , find the distance AB.

Q004 Written

For two points A (4) and B (10), find the distance AB.

Classified Practice

Coordinate Geometry Basics

Q005 MCQ

Find a (-4) , B (8) : distance AB:**Choose the correct option.**

A 11

B -12

C 12

D 13

Q006 MCQ**Find a (-5) , B (7) : distance AB:****Choose the correct option.**

A 11

B -12

C 12

D 13

Classified Practice

Coordinate Geometry Basics

Q007 MCQ

For two points A (-2) and B (6) , find the distance AB:**Choose the correct option.**

A 7

B -8

C 8

D 9

Q008 MCQ**For two points A (4) and B (10), find the distance AB:****Choose the correct option.**

A 5

B -6

C 6

D 7

Classified Practice

Coordinate Geometry Basics

Q009 Written

A (−4), B (8): internal point C in ratio 2:1

Q010 Written**A (-5) , B (7) : internal point C in ratio 2:1**

Classified Practice

Coordinate Geometry Basics

Q011 **Written**

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1.

Q012 Written

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1.

Classified Practice

Coordinate Geometry Basics

Q013 MCQ

Find a (-4) , B (8) : internal point C in ratio 2:1:**Choose the correct option.**A -4 B 4 C 5 D 3

Q014 MCQ**Find a (-5) , B (7) : internal point C in ratio 2:1:****Choose the correct option.****A** 2**B** -3 **C** 3**D** 4

Classified Practice

Coordinate Geometry Basics

Q015 MCQ

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1:

Choose the correct option.

A -4

B 4

C 5

D 3

Q016 MCQ

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1:

Choose the correct option.

A -8

B 8

C 9

D 7

Classified Practice

Coordinate Geometry Basics

Q017 **Written****A (−4), B (8): external point D in ratio 2:3**

Q018 Written**A (−5), B (7): external point D in ratio 2:5**

Classified Practice

Coordinate Geometry Basics

Q019 **Written**

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3.

Q020 Written

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3.

Classified Practice

Coordinate Geometry Basics

Q021 MCQ

Find a (-4) , B (8) : external point D in ratio 2:3:**Choose the correct option.**A -28 B -27 C -29 D 28

Q022 MCQ**Find a (-5) , B (7) : external point D in ratio 2:5:****Choose the correct option.****A** 13**B** -13 **C** -12 **D** -14

Classified Practice

Coordinate Geometry Basics

Q023 MCQ

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3:

Choose the correct option.

A 18

B 19

C 17

D -18

Q024 MCQ

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3:

Choose the correct option.

A -1

B 1

C 2

D 0

Classified Practice

Coordinate Geometry Basics

Q025 Written

A (−4), B (8): midpoint M

Q026 Written**A (−5), B (7): midpoint M**

Classified Practice

Coordinate Geometry Basics

Q027 **Written**

For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB.

Q028 Written

For two points A (4) and B (10), find the coordinate of midpoint M of segment AB.

Classified Practice

Coordinate Geometry Basics

Q029 MCQ

Find a (-4) , B (8) : midpoint M:**Choose the correct option.**

A 1

B -2

C 2

D 3

Q030 MCQ**Find a (-5) , B (7) : midpoint M:****Choose the correct option.****A 1****B 2****C 0****D -1**

Classified Practice

Coordinate Geometry Basics

Q031 MCQ

For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB:

Choose the correct option.

A 3

B 1

C -2

D 2

Q032 MCQ**For two points A (4) and B (10), find the coordinate of midpoint M of segment AB:****Choose the correct option.****A** 7**B** 8**C** 6**D** -7

Classified Practice

Coordinate Geometry Basics

Q033 **Challenge****Pods A (-8) and B (-3) : fibre-optic tether length**

Classified Practice

Coordinate Geometry Basics

Q034 **Challenge****Pods A (-8) , B (-3) : internal sensor point in ratio 1:4**

Classified Practice

Coordinate Geometry Basics

Q035 **Challenge****Pods A (-8) , B (-3) : external outpost in ratio 3:4**

Classified Practice

Coordinate Geometry Basics

Q036 **Challenge****Pods A (-8) , B (-3) : midpoint supply hub**

Classified Practice

Coordinate Geometry Basics

Q037 **Challenge****Find pods A (-8) and B (-3) : fibre-optic tether length:****Choose the correct option.**

A 5

B 6

C 4

D -5

Classified Practice

Coordinate Geometry Basics

Q038 **Challenge****Find pods A (-8) , B (-3) : internal sensor point in ratio 1:4:****Choose the correct option.**A -8 B 7 C -7 D -6

Classified Practice

Coordinate Geometry Basics

Q039 **Challenge****Find pods A (-8) , B (-3) : external outpost in ratio 3:4:****Choose the correct option.**

A 23

B -23 C -22 D -24

Classified Practice

Coordinate Geometry Basics

Q040 Challenge**Find pods A (-8) , B (-3) : midpoint supply hub:****Choose the correct option.**A $-13/2$ B $11/2$ C $-11/2$ D $-9/2$

Classified Practice

Coordinate Geometry Basics

Q041 Written

Point a has coordinates (2,3): determine its quadrant

Q042 Written**Point b has coordinates (2,-3): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q043 Written

Point c has coordinates (-2,3): determine its quadrant

Q044 Written**Point d has coordinates $(-2, -3)$: determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q045 Written

Point a has coordinates $(-4,2)$: determine its quadrant

Q046 Written**Point b has coordinates (1,2): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q047 **Written****Point c has coordinates (3,-2): determine its quadrant**

Q048 Written**Point d has coordinates $(-3,-1)$: determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q049 **Written**

Point a has coordinates (-3,1): determine its quadrant

Q050 Written**Point b has coordinates (2,-8): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q051 Written

Point c has coordinates (4,4): determine its quadrant

Q052 Written**Point d has coordinates $(-6, -4)$: determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q053 Written

Point e has coordinates (-2,6): determine its quadrant

Q054 Written**Point f has coordinates (3,-3): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q055 Written

Point g has coordinates $(-5, -7)$: determine its quadrant

Q056 Written**Point h has coordinates (1,4): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q057 MCQ

Point a has coordinates (2,3): determine its quadrant:**Choose the correct option.**A *III*B *IV*C *I*D *II*

Q058 MCQ**Point b has coordinates (2,-3): determine its quadrant:****Choose the correct option.****A** *III***B** *IV***C** *I***D** *II*

Classified Practice

Coordinate Geometry Basics

Q059 MCQ

Point c has coordinates (-2,3): determine its quadrant:**Choose the correct option.**A *II*B *I*C *III*D *IV*

Q060 MCQ**Point d has coordinates $(-2,-3)$: determine its quadrant:****Choose the correct option.****A** *II***B** *IV***C** *III***D** *I*

Classified Practice

Coordinate Geometry Basics

Q061 MCQ

Point a has coordinates (-4,2): determine its quadrant:**Choose the correct option.**A *II*B *I*C *III*D *IV*

Q062 MCQ**Point b has coordinates (1,2): determine its quadrant:****Choose the correct option.****A** *III***B** *IV***C** *I***D** *II*

Classified Practice

Coordinate Geometry Basics

Q063 MCQ

Point c has coordinates (3,-2): determine its quadrant:**Choose the correct option.**A *III*B *IV*C *I*D *II*

Q064 MCQ**Point d has coordinates $(-3,-1)$: determine its quadrant:****Choose the correct option.****A** *III***B** *I***C** *II***D** *IV*

Classified Practice

Coordinate Geometry Basics

Q065 MCQ

Point a has coordinates (-3,1): determine its quadrant:**Choose the correct option.**A *IV*B *II*C *I*D *III*

Q066 MCQ**Point b has coordinates (2,-8): determine its quadrant:****Choose the correct option.****A** *IV***B** *I***C** *II***D** *III*

Classified Practice

Coordinate Geometry Basics

Q067 MCQ

Point c has coordinates (4,4): determine its quadrant:**Choose the correct option.**A *I*B *II*C *III*D *IV*

Q068 MCQ**Point d has coordinates $(-6, -4)$: determine its quadrant:****Choose the correct option.****A** *II***B** *IV***C** *III***D** *I*

Classified Practice

Coordinate Geometry Basics

Q069 MCQ

Point e has coordinates (-2,6): determine its quadrant:**Choose the correct option.**A *IV*B *II*C *I*D *III*

Q070 MCQ**Point f has coordinates (3,-3): determine its quadrant:****Choose the correct option.****A** *II***B** *III***C** *IV***D** *I*

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Coordinate Geometry Basics

Q071 MCQ

Point g has coordinates (-5,-7): determine its quadrant:**Choose the correct option.**A *II*B *IV*C *III*D *I*

Q072 MCQ**Point h has coordinates (1,4): determine its quadrant:****Choose the correct option.****A** *II***B** *III***C** *IV***D** *I*

Classified Practice

Coordinate Geometry Basics

Q073 Written

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Q074 Written

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Classified Practice

Coordinate Geometry Basics

Q075 **Written**

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Q076 Written

For point A (1, -2), find the coordinates of point P symmetric with respect to the x-axis.

Classified Practice

Coordinate Geometry Basics

Q077 **Written**

For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis.

Q078 MCQ**For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis:****Choose the correct option.****A $(-3, 2)$** **B $(3, -2)$** **C $(-3, -2)$** **D $(-2, -3)$**

Classified Practice

Coordinate Geometry Basics

Q079 MCQ

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Choose the correct option.

A $(-2, -1)$

B $(-1, 2)$

C $(1, -2)$

D $(-1, -2)$

Q080 MCQ**For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis:****Choose the correct option.****A $(-4, -2)$** **B $(4, 2)$** **C $(-4, 2)$** **D $(2, -4)$**

Classified Practice

Coordinate Geometry Basics

Q081 MCQ

For point A $(1, -2)$, find the coordinates of point P symmetric with respect to the x-axis:

Choose the correct option.

A $(2, 1)$

B $(1, -2)$

C $(-1, 2)$

D $(1, 2)$

Q082 MCQ**For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis:****Choose the correct option.****A $(-3, 4)$** **B $(3, -4)$** **C $(-3, -4)$** **D $(-4, -3)$**

Classified Practice

Coordinate Geometry Basics

Q083 **Written**

For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Q084 Written

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Classified Practice

Coordinate Geometry Basics

Q085 **Written**

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Q086 Written

For point A (1, -2), find the coordinates of point Q symmetric with respect to the y-axis.

Classified Practice

Coordinate Geometry Basics

Q087 Written

For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis.

Q088 MCQ**For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:****Choose the correct option.**A $(3, -2)$ B $(-3, 2)$ C $(3, 2)$ D $(2, 3)$

Classified Practice

Coordinate Geometry Basics

Q089 MCQ

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Choose the correct option.

A $(1, -2)$

B $(-1, 2)$

C $(1, 2)$

D $(2, 1)$

Q090 MCQ**For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:****Choose the correct option.****A $(4, -2)$** **B $(-2, 4)$** **C $(4, 2)$** **D $(-4, -2)$**

Classified Practice

Coordinate Geometry Basics

Q091 MCQ

For point A $(1, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Choose the correct option.

A $(-1, -2)$

B $(-2, -1)$

C $(-1, 2)$

D $(1, -2)$

Q092 MCQ**For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis:****Choose the correct option.****A $(3, 4)$** **B $(4, 3)$** **C $(3, -4)$** **D $(-3, 4)$**

Classified Practice

Coordinate Geometry Basics

Q093 Written

For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin.

Q094 Written

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin.

Classified Practice

Coordinate Geometry Basics

Q095 Written

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin.

Q096 Written

For point A (1, -2), find the coordinates of point R symmetric with respect to the origin.

Classified Practice

Coordinate Geometry Basics

Q097 **Written**

For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin.

Q098 MCQ**For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin:****Choose the correct option.**A $(-3, -2)$ B $(3, -2)$ C $(-2, 3)$ D $(3, 2)$

Classified Practice

Coordinate Geometry Basics

Q099 MCQ

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-1, -2)$

B $(1, -2)$

C $(-2, 1)$

D $(1, 2)$

Q100 MCQ

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-4, 2)$

B $(4, 2)$

C $(2, 4)$

D $(4, -2)$

Classified Practice

Coordinate Geometry Basics

Q101 MCQ

For point A $(1, -2)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-1, 2)$

B $(2, -1)$

C $(-1, -2)$

D $(1, 2)$

Q102 MCQ

For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-4, 3)$

B $(3, 4)$

C $(-3, -4)$

D $(3, -4)$

Classified Practice

Coordinate Geometry Basics

Q103 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis.

Classified Practice

Coordinate Geometry Basics

Q104 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis.

Classified Practice

Coordinate Geometry Basics

Q105 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin.

Classified Practice

Coordinate Geometry Basics

Q106 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis:

Choose the correct option.

A $(-3, -4)$

B $(-4, -3)$

C $(-3, 4)$

D $(3, -4)$

Classified Practice

Coordinate Geometry Basics

Q107 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis:

Choose the correct option.

A $(-3, 4)$

B $(3, 4)$

C $(4, 3)$

D $(3, -4)$

Classified Practice

Coordinate Geometry Basics

Q108 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(3, 4)$

B $(-3, -4)$

C $(3, -4)$

D $(-4, 3)$

Classified Practice

Coordinate Geometry Basics

Q109 **Written**

Distance between A (2, 6) and B (5, 10)

Q110 Written**Distance between A (4, 3) and B (1, 2)**

Classified Practice

Coordinate Geometry Basics

Q111 Written

Distance between $A(\sqrt{3}, -1)$ and $B(0, 1)$

Q112 Written**Distance between A (1, 2) and B (11, 7)**

Classified Practice

Coordinate Geometry Basics

Q113 **Written****Distance between A (3, -1) and B (-1, 3)**

Q114 MCQ**Find distance between A (2, 6) and B (5, 10):****Choose the correct option.**

A 4

B -5

C 5

D 6

Classified Practice

Coordinate Geometry Basics

Q115 MCQ

Find distance between A (4, 3) and B (1, 2):**Choose the correct option.**

A $\sqrt{10}$

B $2\sqrt{10}$

C (10)

D 1

Q116 MCQ

Find distance between A($\sqrt{3}$, -1) and B (0, 1):**Choose the correct option.**

A (7)

B 1

C $\sqrt{7}$ D $2\sqrt{7}$

Classified Practice

Coordinate Geometry Basics

Q117 MCQ

Find distance between A (1, 2) and B (11, 7):**Choose the correct option.**

A 1

B $5\sqrt{5}$ C $52\sqrt{5}$ D $5(5)$

Q118 MCQ**Find distance between A (3, -1) and B (-1, 3):****Choose the correct option.**

A $4\sqrt{2}$

B $42\sqrt{2}$

C $4(2)$

D 1

Classified Practice

Coordinate Geometry Basics

Q119 **Written****A (3, -2), B (x, 1), distance 5: find x**

Q120 Written

A $(-1, 3)$, **B** $(x, 1)$, distance $\sqrt{5}$: find **x**

Classified Practice

Coordinate Geometry Basics

Q121 **Written****A (3, -1), B (x, 2), distance 5: find x**

Q122 Written

A (1, -3), **B** (-2, y), **distance** $\sqrt{13}$: **find** y

Classified Practice

Coordinate Geometry Basics

Q123 Written

A $(-4, -2)$, **B** $(x, 3)$, distance $5\sqrt{5}$: find **x**

Q124 MCQ

A $(3, -2)$, **B** $(x, 1)$, **distance 5: find x:**

Choose the correct option.

A $x = -1, 7$ (one value only)

B $x = -1, 7$

C $x = +1, 7$

D $x = -17$

Classified Practice

Coordinate Geometry Basics

Q125 MCQ

A $(-1, 3)$, B $(x, 1)$, distance $\sqrt{5}$: find x:

Choose the correct option.

A $x = -20$

B $x = -2, 0$ (one value only)

C $x = -2, 0$

D $x = +2, 0$

Q126 MCQ**A (3, -1), B (x, 2), distance 5: find x:****Choose the correct option.****A** $x = -1, 7$ (one value only)**B** $x = -1, 7$ **C** $x = +1, 7$ **D** $x = -17$

Classified Practice

Coordinate Geometry Basics

Q127 MCQ

A (1, -3), B (-2, y), distance $\sqrt{13}$: find y:

Choose the correct option.

A $y = -5, -1$

B $y = +5, -1$

C $y = -5 - 1$

D $y = -5, -1$ (one value only)

Q128 MCQ**A** $(-4, -2)$, **B** $(x, 3)$, **distance** $5\sqrt{5}$: **find x:****Choose the correct option.****A** $x = +14, 6$ **B** $x = -146$ **C** $x = -14, 6$ (one value only)**D** $x = -14, 6$

Classified Practice

Coordinate Geometry Basics

Q129 **Written****Distance from origin to A (2, -4)**

Q130 Written**Distance from origin to A (−3, 4)**

Classified Practice

Coordinate Geometry Basics

Q131 MCQ

Find distance from origin to A (2, -4):**Choose the correct option.**A $2(5)$

B 1

C $2\sqrt{5}$ D $22\sqrt{5}$

Q132 MCQ

Find distance from origin to A $(-3, 4)$:**Choose the correct option.**

A 4

B -5

C 5

D 6

Classified Practice

Coordinate Geometry Basics

Q133 **Challenge****Distance between $(x, 7)$ and $(3x - 1, -5)$ is 13: find x**

Classified Practice

Coordinate Geometry Basics

Q134 Challenge

Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x :

Choose the correct option.

A $x = -23$

B $x = -2, 3$ (one value only)

C $x = -2, 3$

D $x = +2, 3$

Classified Practice

Coordinate Geometry Basics

Q135 **Written****Point on the x-axis equidistant from A (−2, 1) and B (3, 4)**

Q136 Written**Point on the y-axis equidistant from A (4, 2) and B (1, -1)**

Classified Practice

Coordinate Geometry Basics

Q137 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(4, 3)$**

Q138 Written**Point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$**

Classified Practice

Coordinate Geometry Basics

Q139 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$**

Q140 MCQ**Find point on the x-axis equidistant from A $(-2, 1)$ and B $(3, 4)$:****Choose the correct option.**A $(-2, 0)$ B $(-3, 0)$ C $(2, 0)$ D $(0, 2)$

Classified Practice

Coordinate Geometry Basics

Q141 MCQ

Find point on the y-axis equidistant from A (4, 2) and B (1, -1):**Choose the correct option.**

A (0, 3)

B (3, 0)

C (0, -3)

D (-1, 3)

Q142 MCQ**Find point on the x-axis equidistant from A $(-1, 2)$ and B $(4, 3)$:****Choose the correct option.**A $(-2, 0)$ B $(-3, 0)$ C $(2, 0)$ D $(0, 2)$

Classified Practice

Coordinate Geometry Basics

Q143 MCQ

Find point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$:**Choose the correct option.**A $(0, -5/6)$ B $(-1, 5/6)$ C $(0, 5/6)$ D $(5/6, 0)$

Q144 MCQ**Find point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$:****Choose the correct option.**A $(-19/2, 0)$ B $(17/2, 0)$ C $(0, 17/2)$ D $(-17/2, 0)$

Classified Practice

Coordinate Geometry Basics

Q145 **Written****Point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1)**

Q146 Written**Point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3)**

Classified Practice

Coordinate Geometry Basics

Q147 **Written****Point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2)**

Q148 Written**Point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$**

Classified Practice

Coordinate Geometry Basics

Q149 Written

Point on $y = -x + 1$ equidistant from A (1, 2) and B (−3, 6)

Q150 MCQ**Find point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1):****Choose the correct option.****A** $(-13/8, 9/4)$ **B** $(13/8, 9/4)$ **C** $(9/4, 13/8)$ **D** $(13/8, -9/4)$

Classified Practice

Coordinate Geometry Basics

Q151 MCQ

Find point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3):**Choose the correct option.**

A (-4, 11)

B (4, -11)

C (-4, -11)

D (-11, -4)

Q152 MCQ**Find point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2):****Choose the correct option.**

A (2, 1)

B (1, 2)

C (2, -1)

D (-2, 1)

Classified Practice

Coordinate Geometry Basics

Q153 MCQ

Find point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$:**Choose the correct option.**A $(3, -2)$ B $(-3, 2)$ C $(3, 2)$ D $(2, 3)$

Q154 MCQ**Find point on $y = -x + 1$ equidistant from A (1, 2) and B (−3, 6):****Choose the correct option.****A** (2, 3)**B** (−2, 3)**C** (3, −2)**D** (−2, −3)

Classified Practice

Coordinate Geometry Basics

Q155 **Challenge**

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD

Classified Practice

Coordinate Geometry Basics

Q156 Challenge

A (2, 1), B (4, 5), C (3, y) with $CA=CB$ and D (x, 0) with $DA=DB$: find CD:

Choose the correct option.

A $3\sqrt{5}$

B $32\sqrt{5}$

C 3(5)

D 1

Classified Practice

Coordinate Geometry Basics

Q157 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Classified Practice

Coordinate Geometry Basics

Q158 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1.

Classified Practice

Coordinate Geometry Basics

Q159 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1.

Classified Practice

Coordinate Geometry Basics

Q160 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2.

Classified Practice

Coordinate Geometry Basics

Q161 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Classified Practice

Coordinate Geometry Basics

Q162 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Choose the correct option.

A $(4/3, -11/3)$

B $(-4/3, 11/3)$

C $(4/3, 11/3)$

D $(11/3, 4/3)$

Classified Practice

Coordinate Geometry Basics

Q163 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1:

Choose the correct option.

A $(-15/4, -13/4)$

B $(15/4, 13/4)$

C $(-15/4, 13/4)$

D $(13/4, -15/4)$

Classified Practice

Coordinate Geometry Basics

Q164 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1:

Choose the correct option.

A (2, 9/5)

B (-2, -9/5)

C (2, -9/5)

D (-9/5, 2)

Classified Practice

Coordinate Geometry Basics

Q165 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2:

Choose the correct option.

A $(-3, 3)$

B $(-3, -3)$

C $(3, 3)$

D $(3, -3)$

Classified Practice

Coordinate Geometry Basics

Q166 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Choose the correct option.

A $(-2/3, -5/3)$

B $(-5/3, -2/3)$

C $(-2/3, 5/3)$

D $(2/3, -5/3)$

Classified Practice

Coordinate Geometry Basics

Q167 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Classified Practice

Coordinate Geometry Basics

Q168 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Classified Practice

Coordinate Geometry Basics

Q169 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Classified Practice

Coordinate Geometry Basics

Q170 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2.

Classified Practice

Coordinate Geometry Basics

Q171 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Classified Practice

Coordinate Geometry Basics

Q172 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Choose the correct option.

A $(-2, 19/2)$

B $(2, 19/2)$

C $(19/2, 2)$

D $(2, -19/2)$

Classified Practice

Coordinate Geometry Basics

Q173 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Choose the correct option.

A $(11, 18)$

B $(-11, -18)$

C $(11, -18)$

D $(-18, 11)$

Classified Practice

Coordinate Geometry Basics

Q174 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Choose the correct option.

A (-10, -15)

B (10, -15)

C (-15, 10)

D (10, 15)

Classified Practice

Coordinate Geometry Basics

Q175 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2:

Choose the correct option.

A $(8, -11)$

B $(-8, -11)$

C $(-11, -8)$

D $(-8, 11)$

Classified Practice

Coordinate Geometry Basics

Q176 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Choose the correct option.

A $(0, 15/2)$

B $(-1, -15/2)$

C $(0, -15/2)$

D $(-15/2, 0)$

Classified Practice

Coordinate Geometry Basics

Q177 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA.

Classified Practice

Coordinate Geometry Basics

Q178 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q179 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB.

Classified Practice

Coordinate Geometry Basics

Q180 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q181 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA.

Classified Practice

Coordinate Geometry Basics

Q182 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q183 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB.

Classified Practice

Coordinate Geometry Basics

Q184 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q185 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC.

Classified Practice

Coordinate Geometry Basics

Q186 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q187 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA:

Choose the correct option.

A $(3/2, -1/2)$

B $(-1/2, 3/2)$

C $(3/2, 1/2)$

D $(-3/2, -1/2)$

Classified Practice

Coordinate Geometry Basics

Q188 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A $(-2, 2)$

B $(-2, -2)$

C $(2, 2)$

D $(2, -2)$

Classified Practice

Coordinate Geometry Basics

Q189 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB:

Choose the correct option.

A $(0, 0)$

B $(-1, 0)$

C $(0, -1)$

D $(1, 0)$

Classified Practice

Coordinate Geometry Basics

Q190 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A $(-2, 1)$

B $(1, -2)$

C $(-2, -1)$

D $(2, 1)$

Classified Practice

Coordinate Geometry Basics

Q191 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA:

Choose the correct option.

A (0, 3)

B (3, 0)

C (0, -3)

D (-1, 3)

Classified Practice

Coordinate Geometry Basics

Q192 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A (1, 2/3)

B (2/3, -1)

C (-2/3, 1)

D (2/3, 1)

Classified Practice

Coordinate Geometry Basics

Q193 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB:

Choose the correct option.

A $(-2, -1)$

B $(2, -1)$

C $(-1, 2)$

D $(2, 1)$

Classified Practice

Coordinate Geometry Basics

Q194 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A $(5/3, 5/3)$

B $(5/3, -5/3)$

C $(-5/3, 5/3)$

D $(-5/3, -5/3)$

Classified Practice

Coordinate Geometry Basics

Q195 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC:

Choose the correct option.

A $(-2, -5/2)$

B $(2, -5/2)$

C $(-5/2, 2)$

D $(2, 5/2)$

Classified Practice

Coordinate Geometry Basics

Q196 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A 1

B 2

C $(0, 0)$ D $(-1, 0)$

Classified Practice

Coordinate Geometry Basics

Q197 **Challenge****A $(-2, 3)$, B $(6, 1)$, C $(0, y)$, $CA=CB$: find the centroid G**

Classified Practice

Coordinate Geometry Basics

Q198 Challenge

A $(-2, 3)$, **B** $(6, 1)$, **C** $(0, y)$, **CA=CB**: find the centroid **G**:

Choose the correct option.

A $(4/3, 2/3)$

B $(-4/3, -2/3)$

C $(4/3, -2/3)$

D $(-2/3, 4/3)$

Classified Practice

Coordinate Geometry Basics

Q199 **Written****Point with A $(-3, 2)$ symmetric about centre $(2, 6)$**

Q200 Written**Point with A (1, -2) symmetric about centre (2, 3)**

Classified Practice

Coordinate Geometry Basics

Q201 **Written****Point with A (3, 2) symmetric about centre (-1, 5)**

Q202 Written**Point with A $(-3, 1)$ symmetric about centre $(2, 0)$**

Classified Practice

Coordinate Geometry Basics

Q203 **Written**

Point with A (7, 4) symmetric about centre (-3, 5)

Q204 Written**Point with A (4, -8) symmetric about centre (-3, 2)**

Classified Practice

Coordinate Geometry Basics

Q205 **Written****Point with A (0, -4) symmetric about centre (-4, 1)**

Q206 Written**Point with A $(-4, 2)$ symmetric about centre $(-1, -3)$**

Classified Practice

Coordinate Geometry Basics

Q207 MCQ

Find point with A $(-3, 2)$ symmetric about centre $(2, 6)$:**Choose the correct option.**A $(7, -10)$ B $(-7, 10)$ C $(7, 10)$ D $(10, 7)$

Q208 MCQ**Find point with A (1, -2) symmetric about centre (2, 3):****Choose the correct option.**

A (-3, 8)

B (3, 8)

C (8, 3)

D (3, -8)

Classified Practice

Coordinate Geometry Basics

Q209 MCQ

Find point with A (3, 2) symmetric about centre (-1, 5):**Choose the correct option.**

A (-5, 8)

B (8, -5)

C (-5, -8)

D (5, 8)

Q210 MCQ**Find point with A $(-3, 1)$ symmetric about centre $(2, 0)$:****Choose the correct option.**A $(7, 1)$ B $(-7, -1)$ C $(7, -1)$ D $(-1, 7)$

Classified Practice

Coordinate Geometry Basics

Q211 MCQ

Find point with A (7, 4) symmetric about centre (-3, 5):**Choose the correct option.**

A (-13, -6)

B (13, 6)

C (-13, 6)

D (6, -13)

Q212 MCQ**Find point with A (4, -8) symmetric about centre (-3, 2):****Choose the correct option.**

A (-10, -12)

B (10, 12)

C (-10, 12)

D (12, -10)

Classified Practice

Coordinate Geometry Basics

Q213 MCQ

Find point with A (0, -4) symmetric about centre (-4, 1):**Choose the correct option.**

A (8, 6)

B (-8, 6)

C (6, -8)

D (-8, -6)

Q214 MCQ**Find point with A $(-4, 2)$ symmetric about centre $(-1, -3)$:****Choose the correct option.**A $(2, -8)$ B $(-8, 2)$ C $(2, 8)$ D $(-2, -8)$

Classified Practice

Coordinate Geometry Basics

Q215 **Challenge****B (m , 5), C (4, n) symmetric about A (1, -1): find $m+n$**

Classified Practice

Coordinate Geometry Basics

Q216 Challenge

B (m , 5), **C** (4, n) symmetric about **A** (1, -1): find $m+n$:

Choose the correct option.

A -10

B 9

C -9

D -8

Classified Practice

Coordinate Geometry Basics

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Point A $(-3, 4)$ lies in which quadrant?:

Choose the correct option.

A *II*

B *I*

C *III*

D *IV*

Quiz MCQ 2 Concept Quiz · MCQ

Find the distance between A (1, 2) and B (4, 6):

Choose the correct option.

A -5

B 5

C 6

D 4

Classified Practice

Coordinate Geometry Basics

Quiz MCQ 3 [Concept Quiz](#) · MCQ

Point P divides the segment from A $(-2, 3)$ to B $(6, 1)$ internally in ratio 1:3. Find P:

Choose the correct option.

A $(0, -5/2)$

B $(-1, 5/2)$

C $(0, 5/2)$

D $(5/2, 0)$

Quiz MCQ 4 Concept Quiz · MCQ

Point B is symmetric to A (2, -1) about P (-3, 4). Find B:

Choose the correct option.

A (-8, -9)

B (8, 9)

C (-8, 9)

D (9, -8)

Classified Practice

Coordinate Geometry Basics

Quiz WRITTEN 5 Concept Quiz · Written

Find the point on $y = 2x + 1$ equidistant from A (0, 2) and B (4, 0).

Quiz WRITTEN 6 Concept Quiz · Written

For points A (-5) and B (7) , find the point dividing AB externally in ratio 2:5.

Classified Practice

Coordinate Geometry Basics

Quiz WRITTEN 7 Concept Quiz · Written

If the distance between A (3, -1) and B (x, 2) is 5, find x.

Quiz WRITTEN 8 Concept Quiz · Written

Find the centroid of A $(-2, 1)$, B $(6, -3)$, C $(1, 7)$.

Answer key

Use the question number shown on each page.

Q001 12	Q027 2	Q053 <i>II</i>	Q079 D
Q002 12	Q028 7	Q054 <i>IV</i>	Q080 C
Q003 8	Q029 C	Q055 <i>III</i>	Q081 D
Q004 6	Q030 A	Q056 <i>I</i>	Q082 C
Q005 C	Q031 D	Q057 C	Q083 (3, 2)
Q006 C	Q032 A	Q058 B	Q084 (1, 2)
Q007 C	Q033 5	Q059 A	Q085 (4, -2)
Q008 C	Q034 -7	Q060 C	Q086 (-1, -2)
Q009 4	Q035 -23	Q061 A	Q087 (3, 4)
Q010 3	Q036 -11/2	Q062 C	Q088 C
Q011 4	Q037 A	Q063 B	Q089 C
Q012 8	Q038 C	Q064 A	Q090 A
Q013 B	Q039 B	Q065 B	Q091 A
Q014 C	Q040 C	Q066 A	Q092 A
Q015 B	Q041 <i>I</i>	Q067 A	Q093 (3, -2)
Q016 B	Q042 <i>IV</i>	Q068 C	Q094 (1, -2)
Q017 -28	Q043 <i>II</i>	Q069 B	Q095 (4, 2)
Q018 -13	Q044 <i>III</i>	Q070 C	Q096 (-1, 2)
Q019 18	Q045 <i>II</i>	Q071 C	Q097 (3, -4)
Q020 1	Q046 <i>I</i>	Q072 D	Q098 B
Q021 A	Q047 <i>IV</i>	Q073 (-3, -2)	Q099 B
Q022 B	Q048 <i>III</i>	Q074 (-1, -2)	Q100 B
Q023 A	Q049 <i>II</i>	Q075 (-4, 2)	Q101 A
Q024 B	Q050 <i>IV</i>	Q076 (1, 2)	Q102 D
Q025 2	Q051 <i>I</i>	Q077 (-3, -4)	Q103 (-3, -4)
Q026 1	Q052 <i>III</i>	Q078 C	Q104 (3, 4)

Q105 (3, -4)

Q106 A

Q107 B

Q108 C

Q109 5

Q110 $\sqrt{10}$ Q111 $\sqrt{7}$ Q112 $5\sqrt{5}$ Q113 $4\sqrt{2}$

Q114 C

Q115 A

Q116 C

Q117 B

Q118 A

Q119 $x = -1, 7$ Q120 $x = -2, 0$ Q121 $x = -1, 7$ Q122 $y = -5, -1$ Q123 $x = -14, 6$

Q124 B

Q125 C

Q126 B

Q127 A

Q128 D

Q129 $2\sqrt{5}$

Q130 5

Q131 C

Q132 C

Q133 $x = -2, 3$

Q134 C

Q135 (2, 0)

Q136 (0, 3)

Q137 (2, 0)

Q138 (0, 5/6)

Q139 (17/2, 0)

Q140 C

Q141 A

Q142 C

Q143 C

Q144 B

Q145 (13/8, 9/4)

Q146 (-4, -11)

Q147 (2, 1)

Q148 (3, 2)

Q149 (-2, 3)

Q150 B

Q151 C

Q152 A

Q153 C

Q154 B

Q155 $3\sqrt{5}$

Q156 A

Q157 (4/3, 11/3)

Q158 (-15/4, 13/4)

Q159 (2, -9/5)

Q160 (3, 3)

Q161 (-2/3, -5/3)

Q162 C

Q163 C

Q164 C

Q165 C

Q166 A

Q167 (2, 19/2)

Q168 (11, -18)

Q169 (10, -15)

Q170 (-8, -11)

Q171 (0, -15/2)

Q172 B

Q173 C

Q174 B

Q175 B

Q176 C

Q177 (3/2, -1/2)

Q178 (2, 2)

Q179 (-1, 0)

Q180 (-2, 1)

Q181 (0, 3)

Q182 (2/3, 1)

Q183 (2, -1)

Q184 (5/3, 5/3)

Q185 (2, -5/2)

Q186 (0, 0)

Q187 A

Q188 C

Q189 B

Q190 A

Q191 A

Q192 D

Q193 B

Q194 A

Q195 B

Q196 C

Q197 (4/3, -2/3)

Q198 C

Q199 (7, 10)

Q200 (3, 8)

Q201 (-5, 8)

Q202 (7, -1)

Q203 (-13, 6)

Q204 (-10, 12)

Q205 (-8, 6)

Q206 (2, -8)

Q207 C

Q208 B

Q209 A

Q210 C

Q211 C

Q212 C

Q213 B

Q214 A

Q215 -9

Q216 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 C

Quiz WRITTEN 5 (1, 3)

Quiz WRITTEN 6 -13

Quiz WRITTEN 7 $x = -1, 7$

Quiz WRITTEN 8 (5/3, 5/3)

Concept 2 | Triangles and Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Line

$$y - y_1 = m(x - x_1)$$

Perpendicular

$$m_1 m_2 = -1$$

Area

$$A = \frac{1}{2}bh$$

Classified Practice

Triangles and Lines

Q001 **Written**

Determine the shape of triangle ABC with A (3, 2), B (−1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$.

Classified Practice

Triangles and Lines

Q002 **Written**

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$.

Classified Practice

Triangles and Lines

Q003 MCQ

Determine the shape of triangle ABC with A (3, 2), B (−1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$:

Choose the correct option.

A *isosceles (not right – angled)*

B *scalene (not right – angled)*

C *scalene right – angled*

D *equilateral*

Classified Practice

Triangles and Lines

Q004 MCQ

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$:

Choose the correct option.

A *scalene (not right – angled)*

B *scalene right – angled*

C *equilateral*

D *isosceles (not right – angled)*

Q005 Written**Determine the shape of triangle ABC with A $(-1, 2)$, B $(1, 1)$, C $(0, 4)$.**

Classified Practice

Triangles and Lines

Q006 Written

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1).

Q007 MCQ**Determine the shape of triangle ABC with A $(-1, 2)$, B $(1, 1)$, C $(0, 4)$:****Choose the correct option.****A** *scalene right – angled***B** *isosceles right – angled; right angle at A***C** *equilateral***D** *isosceles (not right – angled)*

Classified Practice

Triangles and Lines

Q008 MCQ

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1):

Choose the correct option.

A *equilateral*

B *isosceles (not right – angled)*

C *scalene right – angled*

D *isosceles right – angled; right angle at A*

Q009 Written**Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4).**

Classified Practice

Triangles and Lines

Q010 MCQ

Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4):

Choose the correct option.

A *equilateral*

B *isosceles (not right – angled)*

C *scalene (not right – angled)*

D *scalene right – angled; right angle at A*

Q011 Written**Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2).**

Classified Practice

Triangles and Lines

Q012 **Written****Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4).**

Q013 Written

Determine the shape of triangle ABC with A (−1, 1), B (4, 2), C (1, 4).

Classified Practice

Triangles and Lines

Q014 MCQ

Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2):**Choose the correct option.**A *scalene right – angled*B *isosceles right – angled; right angle at B*C *equilateral*D *isosceles (not right – angled)*

Q015 MCQ**Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4):****Choose the correct option.****A** *isosceles (not right – angled)***B** *scalene right – angled***C** *isosceles right – angled; right angle at B***D** *equilateral*

Classified Practice

Triangles and Lines

Q016 MCQ

Determine the shape of triangle ABC with A $(-1, 1)$, B $(4, 2)$, C $(1, 4)$:

Choose the correct option.

A *scalene right – angled*

B *isosceles right – angled; right angle at C*

C *equilateral*

D *isosceles (not right – angled)*

Classified Practice

Triangles and Lines

Q017 Challenge

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k.

Classified Practice

Triangles and Lines

Q018 Challenge

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k:

Choose the correct option.

A $k = 1$

B $k = -2$

C $k = 2$

D $k = 3$

Classified Practice

Triangles and Lines

Q019 Written

Find the equation of the line through (3,-4) parallel to the x-axis.

Q020 Written**Find the equation of the line through $(-3,8)$ parallel to the y -axis.**

Classified Practice

Triangles and Lines

Q021 Written

Find the equation of the line through (2,6) parallel to the x-axis.

Q022 Written**Find the equation of the line through (3,5) perpendicular to the x-axis.**

Classified Practice

Triangles and Lines

Q023 Written

Find the equation of the line through $(-4, 5)$ parallel to x-axis.

Q024 Written**Find the equation of the line through (2,1) parallel to x-axis.**

Classified Practice

Triangles and Lines

Q025 Written

Find the equation of the line through $(-4, -7)$ parallel to x-axis.

Q026 Written**Find the equation of the line through (1,-2) perpendicular to x-axis.**

Classified Practice

Triangles and Lines

Q027 Written

Find the equation of the line through $(-3, -4)$ perpendicular to x-axis.

Q028 Written**Find the equation of the line through (3,-5) perpendicular to x-axis.**

Classified Practice

Triangles and Lines

Q029 Written

Find the equation of the line through (2,-7) parallel to y-axis.

Q030 Written**Find the equation of the line through (4,3) parallel to y-axis.**

Classified Practice

Triangles and Lines

Q031 **Written****Find the equation of the line through $(-6, -2)$ parallel to y-axis.**

Q032 MCQ**Find the equation of the line through (3,-4) parallel to the x-axis:****Choose the correct option.**

A $y = 4$

B $y = -4$

C $y = -3$

D $y = -5$

Classified Practice

Triangles and Lines

Q033 MCQ

Find the equation of the line through (-3,8) parallel to the y-axis:**Choose the correct option.**

A $x = -4$

B $x = 3$

C $x = -3$

D $x = -2$

Q034 MCQ**Find the equation of the line through (2,6) parallel to the x-axis:****Choose the correct option.**

A $y = 7$

B $y = 5$

C $y = -6$

D $y = 6$

Classified Practice

Triangles and Lines

Q035 MCQ

Find the equation of the line through (3,5) perpendicular to the x-axis:**Choose the correct option.**

A $x = 4$

B $x = 2$

C $x = -3$

D $x = 3$

Q036 MCQ**Find the equation of the line through $(-4,5)$ parallel to x-axis:****Choose the correct option.**

A $y = -5$

B $y = 5$

C $y = 6$

D $y = 4$

Classified Practice

Triangles and Lines

Q037 MCQ

Find the equation of the line through (2,1) parallel to x-axis:**Choose the correct option.**

A $y = 2$

B $y = 0$

C $y = -1$

D $y = 1$

Q038 MCQ**Find the equation of the line through $(-4, -7)$ parallel to x-axis:****Choose the correct option.**

A $y = 7$

B $y = -7$

C $y = -6$

D $y = -8$

Classified Practice

Triangles and Lines

Q039 MCQ

Find the equation of the line through (1,-2) perpendicular to x-axis:**Choose the correct option.**

A $x = 0$

B $x = -1$

C $x = 1$

D $x = 2$

Q040 MCQ**Find the equation of the line through (-3,-4) perpendicular to x-axis:****Choose the correct option.**

A $x = -2$

B $x = -4$

C $x = 3$

D $x = -3$

Classified Practice

Triangles and Lines

Q041 MCQ

Find the equation of the line through (3,-5) perpendicular to x-axis:**Choose the correct option.**

A $x = 4$

B $x = 2$

C $x = -3$

D $x = 3$

Q042 MCQ**Find the equation of the line through (2,-7) parallel to y-axis:****Choose the correct option.**

A $x = -2$

B $x = 2$

C $x = 3$

D $x = 1$

Classified Practice

Triangles and Lines

Q043 MCQ

Find the equation of the line through (4,3) parallel to y-axis:**Choose the correct option.**

A $x = 5$

B $x = 3$

C $x = -4$

D $x = 4$

Q044 MCQ**Find the equation of the line through $(-6, -2)$ parallel to y -axis:****Choose the correct option.**

A $x = -6$

B $x = -5$

C $x = -7$

D $x = 6$

Classified Practice

Triangles and Lines

Q045 **Written****Find the equation of the line through $(-3,1)$ and $(-3,2)$.**

Q046 Written**Find the equation of the line through $(-2,3)$ and $(-5,3)$.**

Classified Practice

Triangles and Lines

Q047 **Written****Find the equation of the line through (1,-1) and (4,0).**

Q048 Written**Find the equation of the line through $(-2,-2)$ and $(2,-2)$.**

Classified Practice

Triangles and Lines

Q049 **Written****Find the equation of the line through $(-4,2)$ and $(-4,8)$.**

Q050 Written**Find the equation of the line through $(-2,-6)$ and $(-4,1)$.**

Classified Practice

Triangles and Lines

Q051 **Written****Find the equation of the line through (4,0) and (4,3).**

Q052 Written**Find the equation of the line through $(-1,2)$ and $(-1,6)$.**

Classified Practice

Triangles and Lines

Q053 Written

Find the equation of the line through $(-5,-3)$ and $(-5,2)$.

Q054 Written**Find the equation of the line through $(-2,3)$ and $(-3,3)$.**

Classified Practice

Triangles and Lines

Q055 **Written****Find the equation of the line through (4,-2) and (1,-2).**

Q056 Written**Find the equation of the line through $(-3,6)$ and $(2,6)$.**

Classified Practice

Triangles and Lines

Q057 **Written****Find the equation of the line through (1,2) and (3,10).**

Q058 Written**Find the equation of the line through $(-1,-8)$ and $(4,2)$.**

Classified Practice

Triangles and Lines

Q059 MCQ

Find the equation of the line through (-3,1) and (-3,2):**Choose the correct option.**

A $x = 3$

B $x = -3$

C $x = -2$

D $x = -4$

Q060 MCQ**Find the equation of the line through $(-2,3)$ and $(-5,3)$:****Choose the correct option.**

A $y = 2$

B $y = -3$

C $y = 3$

D $y = 4$

Classified Practice

Triangles and Lines

Q061 MCQ

Find the equation of the line through (1,-1) and (4,0):**Choose the correct option.**

A $y = -x/3 - 4/3$

B $y = 4/3 - x/3$

C $y = x/3 - 4/3 + 1$

D $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$)

Q062 MCQ**Find the equation of the line through (-2,-2) and (2,-2):****Choose the correct option.**

A $y = -2$

B $y = -1$

C $y = -3$

D $y = 2$

Classified Practice

Triangles and Lines

Q063 MCQ

Find the equation of the line through $(-4,2)$ and $(-4,8)$:**Choose the correct option.**

A $x = 4$

B $x = -4$

C $x = -3$

D $x = -5$

Q064 MCQ**Find the equation of the line through (-2,-6) and (-4,1):****Choose the correct option.**

A $7x + 2y + 26 + 1 = 0$

B $7x - 2y + 26 = 0$

C $2(7x + 2y + 26) + 1 = 0$

D $7x + 2y + 26 = 0$

Classified Practice

Triangles and Lines

Q065 MCQ

Find the equation of the line through (4,0) and (4,3):**Choose the correct option.**

A $x = 3$

B $x = -4$

C $x = 4$

D $x = 5$

Q066 MCQ**Find the equation of the line through $(-1,2)$ and $(-1,6)$:****Choose the correct option.**

A $x = -1$

B $x = 0$

C $x = -2$

D $x = 1$

Classified Practice

Triangles and Lines

Q067 MCQ

Find the equation of the line through $(-5,-3)$ and $(-5,2)$:**Choose the correct option.**

A $x = -6$

B $x = 5$

C $x = -5$

D $x = -4$

Q068 MCQ**Find the equation of the line through $(-2,3)$ and $(-3,3)$:****Choose the correct option.**

A $y = 4$

B $y = 2$

C $y = -3$

D $y = 3$

Classified Practice

Triangles and Lines

Q069 MCQ

Find the equation of the line through (4,-2) and (1,-2):**Choose the correct option.**

A $y = -1$

B $y = -3$

C $y = 2$

D $y = -2$

Q070 MCQ**Find the equation of the line through (-3,6) and (2,6):****Choose the correct option.**

A $y = 5$

B $y = -6$

C $y = 6$

D $y = 7$

Classified Practice

Triangles and Lines

Q071 MCQ

Find the equation of the line through (1,2) and (3,10):**Choose the correct option.**

A $y = -4x - 2$

B $y = 4x - 2 + 1$

C 0

D $y = 4x - 2$

Q072 MCQ**Find the equation of the line through (-1,-8) and (4,2):****Choose the correct option.**

A $y = -2x - 6$

B $y = 2x - 6 + 1$

C 0

D $y = 2x - 6$

Classified Practice

Triangles and Lines

Q073 Written

Find the equation of the line through (2,-1) with gradient 3.

Q074 Written**Find the equation of the line through $(-3,4)$ with gradient 2.**

Classified Practice

Triangles and Lines

Q075 **Written****Find the equation of the line through $(-1, 5)$ with gradient -1 .**

Q076 Written**Find the equation of the line through $(-3, -8)$ with gradient 1.**

Classified Practice

Triangles and Lines

Q077 **Written****Find the equation of the line through (2,-1) with gradient -2.**

Q078 MCQ**Find the equation of the line through (2,-1) with gradient 3:****Choose the correct option.**

A $y = -3x - 7$

B $y = 3x - 7 + 1$

C 0

D $y = 3x - 7$

Classified Practice

Triangles and Lines

Q079 MCQ

Find the equation of the line through (-3,4) with gradient 2:**Choose the correct option.**

A $y = 2x + 10 + 1$

B 0

C $y = 2x + 10$

D $y = -2x + 10$

Q080 MCQ**Find the equation of the line through $(-1,5)$ with gradient -1 :****Choose the correct option.**

A $y = x + 4$

B $y = x - 4$

C $y = -x + 4 + 1$

D $y = -x + 4$

Classified Practice

Triangles and Lines

Q081 MCQ

Find the equation of the line through (-3,-8) with gradient 1:**Choose the correct option.**

A $y = x - 5 + 1$

B $y = x - 5$

C $y = -x - 5$

D $y = 5 - x$

Q082 MCQ**Find the equation of the line through (2,-1) with gradient -2:****Choose the correct option.**

A $y = 2x + 3$

B $y = -2x + 3 + 1$

C 0

D $y = -2x + 3$

Classified Practice

Triangles and Lines

Q083 **Challenge**

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B.

Classified Practice

Triangles and Lines

Q084 Challenge

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B:

Choose the correct option.

A $y = 3$

B $y = 1$

C $y = -2$

D $y = 2$

Classified Practice

Triangles and Lines

Q085 Written

A $(2, 3)$, **B** $(4, -1)$, and **C** $(a - 1, 2 - a)$ are collinear. Find **a**.

Q086 Written

A (2, 5), B (4, 9), and C (−1, a) are collinear. Find a.

Classified Practice

Triangles and Lines

Q087 Written

A (1, -1), **B** (-2, -3), and **C** (4, a) are collinear. Find a .

Q088 **Written**

A (1, 2), B (3, 1), and C (a, -1) are collinear. Find a.

Classified Practice

Triangles and Lines

Q089 Written

A $(-3, -9)$, **B** $(5, 7)$, and **C** $(a + 3, 3a - 2)$ are collinear. Find **a**.

Q090 MCQ**A (2, 3), B (4, -1), and C ($a - 1$, $2 - a$) are collinear. Find a:****Choose the correct option.**

A $a = 8$

B $a = 6$

C $a = -7$

D $a = 7$

Classified Practice

Triangles and Lines

Q091 MCQ

A (2, 5), **B** (4, 9), and **C** (−1, a) are collinear. Find a :

Choose the correct option.

A $a = 0$

B $a = -2$

C $a = 1$

D $a = -1$

Q092 MCQ**A (1, -1), B (-2, -3), and C (4, a) are collinear. Find a:****Choose the correct option.****A $a = 0$** **B $a = -1$** **C $a = 1$** **D $a = 2$**

Classified Practice

Triangles and Lines

Q093 MCQ

A (1, 2), **B** (3, 1), and **C** (a , -1) are collinear. Find a :

Choose the correct option.

A $a = 8$

B $a = 6$

C $a = -7$

D $a = 7$

Q094 MCQ**A** $(-3, -9)$, **B** $(5, 7)$, and **C** $(a + 3, 3a - 2)$ are collinear. Find **a**:**Choose the correct option.**

A $a = 6$

B $a = 4$

C $a = -5$

D $a = 5$

Classified Practice

Triangles and Lines

Q095 Written

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1, 3)$.

Q096 Written**Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6).**

Classified Practice

Triangles and Lines

Q097 Written

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$.

Q098 Written

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$.

Classified Practice

Triangles and Lines

Q099 MCQ

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1, 3)$:

Choose the correct option.

A $2(x + y - 2) + 1 = 0$

B $x + y - 2 = 0$

C $x + y - 2 + 1 = 0$

D $x - y - 2 = 0$

Q100 MCQ

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6):

Choose the correct option.

A $x + 3y - 23 = 0$

B $x + 3y - 23 + 1 = 0$

C $x - 3y - 23 = 0$

D $2(x + 3y - 23) + 1 = 0$

Classified Practice

Triangles and Lines

Q101 MCQ

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$:

Choose the correct option.

A $2(x + y - 1) + 1 = 0$

B $x + y - 1 = 0$

C $x + y - 1 + 1 = 0$

D $x - y - 1 = 0$

Q102 MCQ

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$:

Choose the correct option.

A $2x - 5y + 1 + 1 = 0$

B $2x - 5y - 1 = 0$

C $2(2x - 5y + 1) + 1 = 0$

D $2x - 5y + 1 = 0$

Classified Practice

Triangles and Lines

Q103 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory.

Classified Practice

Triangles and Lines

Q104 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory:

Choose the correct option.

A $3x - 13y = 0$

B $2(3x + 13y) + 1 = 0$

C $3x + 13y = 0$

D $3x + 13y + 1 = 0$

Classified Practice

Triangles and Lines

Q105 Written

Find the gradient of line $y = 2x + 3$.

Q106 Written

Find the gradient of line $y = -3x + 2$.

Classified Practice

Triangles and Lines

Q107 **Written****Find the gradient of line $x - 3y + 1 = 0$.**

Q108 Written**Find the gradient of line $2x - y - 1 = 0$.**

Classified Practice

Triangles and Lines

Q109 Written

Find the gradient of line $x + 3y + 3 = 0$.

Q110 Written**Find the gradient of line $y = 3x + 1$.**

Classified Practice

Triangles and Lines

Q111 Written

Find the gradient of line $y = 4x - 1$.

Q112 Written**Find the gradient of line $y = -4x - 7$.**

Classified Practice

Triangles and Lines

Q113 Written

Find the gradient of line $6x - 2y = 5$.

Q114 Written**Find the gradient of line $8x + 2y - 3 = 0$.**

Classified Practice

Triangles and Lines

Q115 **Written****Find the gradient of line $x + 4y - 4 = 0$.**

Q116 Written**Find the gradient of line $y = 3x - 1$.**

Classified Practice

Triangles and Lines

Q117 **Written****Find the gradient of line $y = -3x + 4$.**

Q118 Written**Find the gradient of line $y = x - 3$.**

Classified Practice

Triangles and Lines

Q119 **Written****Find the gradient of line $6x + 2y + 5 = 0$.**

Q120 Written**Find the gradient of line $2x - 2y - 1 = 0$.**

Classified Practice

Triangles and Lines

Q121 Written

Find the gradient of line $x + 3y - 2 = 0$.

Q122 Written**Find the gradient of line $y = 2x - 1$.**

Classified Practice

Triangles and Lines

Q123 Written

Find the gradient of line $y = -x + 4$.

Q124 Written**Find the gradient of line $y = 3x/2 + 6$.**

Classified Practice

Triangles and Lines

Q125 Written

Find the gradient of line $x + y + 3 = 0$.

Q126 Written**Find the gradient of line $2x + 3y + 8 = 0$.**

Classified Practice

Triangles and Lines

Q127 **Written****Find the gradient of line $3x + 6y - 1 = 0$.**

Q128 MCQ

Find the gradient of line $y = 2x + 3$:**Choose the correct option.**

A $m = 3$

B $m = 1$

C $m = -2$

D $m = 2$

Classified Practice

Triangles and Lines

Q129 MCQ

Find the gradient of line $y = -3x + 2$:**Choose the correct option.**A $m = -4$ B $m = 3$ C $m = -3$ D $m = -2$

Q130 MCQ**Find the gradient of line $x - 3y + 1 = 0$:****Choose the correct option.**

A $m = 1/3$

B $m = 4/3$

C $m = -2/3$

D $m = -1/3$

Classified Practice

Triangles and Lines

Q131 MCQ

Find the gradient of line $2x - y - 1 = 0$:**Choose the correct option.**

A $m = -2$

B $m = 2$

C $m = 3$

D $m = 1$

Q132 MCQ**Find the gradient of line $x + 3y + 3 = 0$:****Choose the correct option.**

A $m = -1/3$

B $m = 2/3$

C $m = -4/3$

D $m = 1/3$

Classified Practice

Triangles and Lines

Q133 MCQ

Find the gradient of line $y = 3x + 1$:**Choose the correct option.**

A $m = 3$

B $m = 4$

C $m = 2$

D $m = -3$

Q134 MCQ

Find the gradient of line $y = 4x - 1$:**Choose the correct option.**

A $m = -4$

B $m = 4$

C $m = 5$

D $m = 3$

Classified Practice

Triangles and Lines

Q135 MCQ

Find the gradient of line $y = -4x - 7$:**Choose the correct option.**

A $m = -3$

B $m = -5$

C $m = 4$

D $m = -4$

Q136 MCQ**Find the gradient of line $6x - 2y = 5$:****Choose the correct option.**

A $m = 3$

B $m = 4$

C $m = 2$

D $m = -3$

Classified Practice

Triangles and Lines

Q137 MCQ

Find the gradient of line $8x + 2y - 3 = 0$:**Choose the correct option.**

A $m = -5$

B $m = 4$

C $m = -4$

D $m = -3$

Q138 MCQ**Find the gradient of line $x + 4y - 4 = 0$:****Choose the correct option.**

A $m = -5/4$

B $m = 1/4$

C $m = -1/4$

D $m = 3/4$

Classified Practice

Triangles and Lines

Q139 MCQ

Find the gradient of line $y = 3x - 1$:**Choose the correct option.**

A $m = 4$

B $m = 2$

C $m = -3$

D $m = 3$

Q140 MCQ**Find the gradient of line $y = -3x + 4$:****Choose the correct option.**

A $m = -3$

B $m = -2$

C $m = -4$

D $m = 3$

Classified Practice

Triangles and Lines

Q141 MCQ

Find the gradient of line $y = x - 3$:**Choose the correct option.**

A $m = 2$

B $m = 0$

C $m = -1$

D $m = 1$

Q142 MCQ**Find the gradient of line $6x + 2y + 5 = 0$:****Choose the correct option.**

A $m = -4$

B $m = 3$

C $m = -3$

D $m = -2$

Classified Practice

Triangles and Lines

Q143 MCQ

Find the gradient of line $2x - 2y - 1 = 0$:**Choose the correct option.**

A $m = 0$

B $m = -1$

C $m = 1$

D $m = 2$

Q144 MCQ**Find the gradient of line $x + 3y - 2 = 0$:****Choose the correct option.**

A $m = -1/3$

B $m = 2/3$

C $m = -4/3$

D $m = 1/3$

Classified Practice

Triangles and Lines

Q145 MCQ

Find the gradient of line $y = 2x - 1$:**Choose the correct option.**

A $m = 3$

B $m = 1$

C $m = -2$

D $m = 2$

Q146 MCQ**Find the gradient of line $y = -x + 4$:****Choose the correct option.**

A $m = -1$

B $m = 0$

C $m = -2$

D $m = 1$

Classified Practice

Triangles and Lines

Q147 MCQ

Find the gradient of line $y = 3x/2 + 6$:**Choose the correct option.**

A $m = -3/2$

B $m = 3/2$

C $m = 5/2$

D $m = 1/2$

Q148 MCQ**Find the gradient of line $x + y + 3 = 0$:****Choose the correct option.**

- A $m = 0$
- B $m = -2$
- C $m = 1$
- D $m = -1$

Classified Practice

Triangles and Lines

Q149 MCQ

Find the gradient of line $2x + 3y + 8 = 0$:**Choose the correct option.**

A $m = 2/3$

B $m = -2/3$

C $m = 1/3$

D $m = -5/3$

Q150 MCQ**Find the gradient of line $3x + 6y - 1 = 0$:****Choose the correct option.****A** 0**B** $m = -1/2$ **C** $m = 1/2$ **D** $m = -3/2$

Classified Practice

Triangles and Lines

Q151 **Written****Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$.**

Q152 Written

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$.

Classified Practice

Triangles and Lines

Q153 **Written****Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$.**

Q154 Written

Find the equation of the line through $(-1,2)$ parallel to $2x + 3y = 0$.

Classified Practice

Triangles and Lines

Q155 **Written**

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$.

Q156 MCQ

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$:**Choose the correct option.**

- A $3x + 2y - 14 = 0$
- B $2(3x - 2y - 14) + 1 = 0$
- C $3x - 2y - 14 = 0$
- D $3x - 2y - 14 + 1 = 0$

Classified Practice

Triangles and Lines

Q157 MCQ

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$:**Choose the correct option.**

- A $2x + y + 1 + 1 = 0$
B $2x - y + 1 = 0$
C $2(2x + y + 1) + 1 = 0$
D $2x + y + 1 = 0$
-
-
-

Q158 MCQ**Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$:****Choose the correct option.**

A $2x + y - 4 = 0$

B $2(2x - y - 4) + 1 = 0$

C $2x - y - 4 = 0$

D $2x - y - 4 + 1 = 0$

Classified Practice

Triangles and Lines

Q159 MCQ

Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$:**Choose the correct option.**

A $2x + 3y - 4 + 1 = 0$

B $2x - 3y - 4 = 0$

C $2(2x + 3y - 4) + 1 = 0$

D $2x + 3y - 4 = 0$

Q160 MCQ

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$:

Choose the correct option.

A $3x + 2y - 7 + 1 = 0$

B $3x - 2y - 7 = 0$

C $2(3x + 2y - 7) + 1 = 0$

D $3x + 2y - 7 = 0$

Classified Practice

Triangles and Lines

Q161 Written

Find the line through $(-2, 5)$ perpendicular to $3x + 5y + 1 = 0$.

Q162 Written

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$.

Classified Practice

Triangles and Lines

Q163 **Written**

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$.

Q164 Written

Find the equation of the line through $(-1,3)$ perpendicular to $5x - 2y + 3 = 0$.

Classified Practice

Triangles and Lines

Q165 Written

Find the equation of the line through $(-2, -5)$ perpendicular to $3x - 2y - 7 = 0$.

Q166 MCQ**Find the line through $(-2,5)$ perpendicular to $3x + 5y + 1 = 0$:****Choose the correct option.**

A $5x - 3y + 25 + 1 = 0$

B $5x - 3y - 25 = 0$

C $2(5x - 3y + 25) + 1 = 0$

D $5x - 3y + 25 = 0$

Classified Practice

Triangles and Lines

Q167 MCQ

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$:**Choose the correct option.**

A $3x - 2y - 4 + 1 = 0$

B $3x + 2y - 4 = 0$

C $2(3x - 2y - 4) + 1 = 0$

D $3x - 2y - 4 = 0$

Q168 MCQ**Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$:****Choose the correct option.**

A $2(x - 2y - 12) + 1 = 0$

B $x - 2y - 12 = 0$

C $x - 2y - 12 + 1 = 0$

D $x + 2y - 12 = 0$

Classified Practice

Triangles and Lines

Q169 MCQ

Find the equation of the line through $(-1,3)$ perpendicular to $5x - 2y + 3 = 0$:

Choose the correct option.

A $2x + 5y - 13 = 0$

B $2x + 5y - 13 + 1 = 0$

C $2x - 5y - 13 = 0$

D $2(2x + 5y - 13) + 1 = 0$

Q170 MCQ

Find the equation of the line through $(-2, -5)$ perpendicular to $3x - 2y - 7 = 0$:

Choose the correct option.

A $2x + 3y + 19 + 1 = 0$

B $2x - 3y + 19 = 0$

C $2(2x + 3y + 19) + 1 = 0$

D $2x + 3y + 19 = 0$

Classified Practice

Triangles and Lines

Q171 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k .

Classified Practice

Triangles and Lines

Q172 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k :

Choose the correct option.

- A $m/k = 1$ ($k = 2, m = 2$)
- B $m/k = 3$ ($k = 2, m = 6$)
- C $m/k = -2$ ($k = 1, m = -2$)
- D $m/k = 2$ ($k = 3, m = 6$)

Classified Practice

Triangles and Lines

Q173 **Written**

For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB.

Q174 Written

For A $(-1, 2)$, B $(3, -4)$, find the equation of the perpendicular bisector of AB.

Classified Practice

Triangles and Lines

Q175 **Written****Find the perpendicular bisector of segment A $(-2, 3)$, B $(2, 1)$.**

Q176 Written**Find the perpendicular bisector of segment A $(-1, 2)$, B $(5, -2)$.**

Classified Practice

Triangles and Lines

Q177 **Written****Find the perpendicular bisector of segment A (0, 6), B (4, 4).**

Q178 MCQ**For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB:****Choose the correct option.**

A $x + 3y - 18 = 0$

B $x + 3y - 18 + 1 = 0$

C $x - 3y - 18 = 0$

D $2(x + 3y - 18) + 1 = 0$

Classified Practice

Triangles and Lines

Q179 MCQ

For A $(-1, 2)$, B $(3, -4)$, find the equation of the perpendicular bisector of AB:

Choose the correct option.

A $2x + 3y - 5 = 0$

B $2(2x - 3y - 5) + 1 = 0$

C $2x - 3y - 5 = 0$

D $2x - 3y - 5 + 1 = 0$

Q180 MCQ**Find the perpendicular bisector of segment A $(-2, 3)$, B $(2, 1)$:****Choose the correct option.**

A $2x - y - 2 = 0$

B $2(2x - y + 2) + 1 = 0$

C $2x - y + 2 = 0$

D $2x - y + 2 + 1 = 0$

Classified Practice

Triangles and Lines

Q181 MCQ

Find the perpendicular bisector of segment A (−1, 2), B (5, −2):**Choose the correct option.**

A $3x - 2y - 6 + 1 = 0$

B $3x + 2y - 6 = 0$

C $2(3x - 2y - 6) + 1 = 0$

D $3x - 2y - 6 = 0$

Q182 MCQ**Find the perpendicular bisector of segment A (0, 6), B (4, 4):****Choose the correct option.**

A $2x - y + 1 = 0$

B $2x - y + 1 + 1 = 0$

C $2x - y - 1 = 0$

D $2(2x - y + 1) + 1 = 0$

Classified Practice

Triangles and Lines

Q183 **Written**

Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$.

Q184 Written**Find the point B symmetric to A (0, 4) with respect to $2x - y - 1 = 0$.**

Classified Practice

Triangles and Lines

Q185 **Written**

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$.

Q186 Written**Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$.**

Classified Practice

Triangles and Lines

Q187 MCQ

Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$:

Choose the correct option.

A $B = (2, 1)$

B $B = (1, 2)$

C $B = (-2, 1)$

D $B = (2, -1)$

Q188 MCQ**Find the point B symmetric to A (0, 4) with respect to $2x - y - 1 = 0$:****Choose the correct option.**

A $B = (4, 2)$

B $B = (2, 4)$

C $B = (-4, 2)$

D $B = (4, -2)$

Classified Practice

Triangles and Lines

Q189 MCQ

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$:

Choose the correct option.

A $B = (-3, 6)$

B $B = (3, -6)$

C $B = (3, 6)$

D $B = (6, 3)$

Q190 MCQ

Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$:

Choose the correct option.

A $B = (2, 2)$

B $B = (-2, -2)$

C $B = (-2, 2)$

D $B = (2, -2)$

Classified Practice

Triangles and Lines

Q191 **Challenge****Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B.**

Classified Practice

Triangles and Lines

Q192 Challenge

Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B:

Choose the correct option.

A $B = (3, 4)$

B $B = (-3, -4)$

C $B = (-3, 4)$

D $B = (4, -3)$

Classified Practice

Triangles and Lines

Q193 Written

Find the distance from $(-2,1)$ to $3x - 2y + 5 = 0$.

Q194 Written**Find the distance from $(3,-2)$ to $x + 2y - 4 = 0$.**

Classified Practice

Triangles and Lines

Q195 **Written****Find the perpendicular distance from $(-3,2)$ to $2x - 3y + 6 = 0$.**

Q196 Written**Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$.**

Classified Practice

Triangles and Lines

Q197 **Written****Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$.**

Q198 Written**Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$.**

Classified Practice

Triangles and Lines

Q199 Written

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$.

Q200 Written**Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$.**

Classified Practice

Triangles and Lines

Q201 **Written****Find the perpendicular distance from (1,2) to $y = 3x + 1$.**

Q202 MCQ

Find the distance from $(-2,1)$ to $3x - 2y + 5 = 0$:**Choose the correct option.**

A $3/\sqrt{13} = 3\sqrt{13}/13$

B $3 \cdot \sqrt{13}/13 + 1$

C $-1 + 3 \cdot \sqrt{13}/13$

D $-3 \cdot \sqrt{13}/13$

Classified Practice

Triangles and Lines

Q203 MCQ

Find the distance from (3,-2) to $x + 2y - 4 = 0$:**Choose the correct option.**

A $1 + \sqrt{5}$

B $-1 + \sqrt{5}$

C $-\sqrt{5}$

D $\sqrt{5}$

Q204 MCQ

Find the perpendicular distance from $(-3,2)$ to $2x - 3y + 6 = 0$:**Choose the correct option.**

A $1 + 6 \cdot \sqrt{13}/13$

B $-1 + 6 \cdot \sqrt{13}/13$

C $-6 \cdot \sqrt{13}/13$

D $6/\sqrt{13} = 6\sqrt{13}/13$

Classified Practice

Triangles and Lines

Q205 MCQ

Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$:**Choose the correct option.**A -6 B 6 C 7 D 5

Q206 MCQ**Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$:****Choose the correct option.**

A $4\sqrt{5} + 1$

B $4\sqrt{5} - 1$

C 0

D $4\sqrt{5}$

Classified Practice

Triangles and Lines

Q207 MCQ

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$:**Choose the correct option.**

A 2

B 0

C -1

D 1

Q208 MCQ

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$:**Choose the correct option.**

- A $\sqrt{13}$
 - B $1 + \sqrt{13}$
 - C $-1 + \sqrt{13}$
 - D $-\sqrt{13}$
-
-
-

Classified Practice

Triangles and Lines

Q209 MCQ

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$:**Choose the correct option.**

A -10

B 10

C 11

D 9

Q210 MCQ**Find the perpendicular distance from (1,2) to $y = 3x + 1$:****Choose the correct option.**

- A $\sqrt{10}/5 + 1$
 - B $-1 + \sqrt{10}/5$
 - C $-\sqrt{10}/5$
 - D $\sqrt{10}/5$
-
-
-

Classified Practice

Triangles and Lines

Q211 **Written****Find the perpendicular length from the origin to $3x - y = 5$.**

Q212 Written**Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$.**

Classified Practice

Triangles and Lines

Q213 MCQ

Find the perpendicular length from the origin to $3x - y = 5$:**Choose the correct option.**

- A $1 + \sqrt{10}/2$
- B $-1 + \sqrt{10}/2$
- C $-\sqrt{10}/2$
- D $\sqrt{10}/2$

Q214 MCQ**Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$:****Choose the correct option.**

A $-\sqrt{10}$

B $\sqrt{10}$

C $1 + \sqrt{10}$

D $-1 + \sqrt{10}$

Classified Practice

Triangles and Lines

Q215 **Challenge****Find the shortest distance from P (−1, 5) to the radiation line $y = x/3 + 2$.**

Classified Practice

Triangles and Lines

Q216 Challenge

Find the shortest distance from P $(-1, 5)$ to the radiation line $y = x/3 + 2$:

Choose the correct option.

- A $\sqrt{10}$
- B $1 + \sqrt{10}$
- C $-1 + \sqrt{10}$
- D $-\sqrt{10}$

Classified Practice

Triangles and Lines

Q217 **Written****Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1).**

Q218 Written

Find the area for A (1, 5), B (−3, −3), C (3, −6).

Classified Practice

Triangles and Lines

Q219 **Written****Find the area for A $(-2, 1)$, B $(2, -1)$, C $(0, 4)$.**

Q220 Written**Find the area for A (0, -6), B (2, -2), C (1, 5).**

Classified Practice

Triangles and Lines

Q221 **Written****Find the area for A $(-2, 4)$, B $(-3, -5)$, C $(5, -1)$.**

Q222 Written**Find the area for A (2, 5), B (0, 1), C (4, 2).**

Classified Practice

Triangles and Lines

Q223 MCQ

Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1):**Choose the correct option.**

A 11

B 9

C −10

D 10

Q224 MCQ**Find the area for A (1, 5), B (−3, −3), C (3, −6):****Choose the correct option.****A** 29**B** −30**C** 30**D** 31

Classified Practice

Triangles and Lines

Q225 MCQ

Find the area for A $(-2, 1)$, B $(2, -1)$, C $(0, 4)$:**Choose the correct option.**

A 8

B 9

C 7

D -8

Q226 MCQ**Find the area for A (0, -6), B (2, -2), C (1, 5):****Choose the correct option.****A** -9**B** 9**C** 10**D** 8

Classified Practice

Triangles and Lines

Q227 MCQ

Find the area for A $(-2, 4)$, B $(-3, -5)$, C $(5, -1)$:**Choose the correct option.**A -34 B 34 C 35 D 33

Q228 MCQ

Find the area for A (2, 5), B (0, 1), C (4, 2):**Choose the correct option.**

A 8

B 6

C -7

D 7

Classified Practice

Triangles and Lines

Q229 **Challenge****Find the area of the wound triangle A $(-2, 1)$, B $(1, 5)$, C $(-21, -21)$.**

Classified Practice

Triangles and Lines

Q230 Challenge

Find the area of the wound triangle A $(-2, 1)$, B $(1, 5)$, C $(-21, -21)$:

Choose the correct option.

A 6

B 4

C -5

D 5

Classified Practice

Triangles and Lines

Q231 **Written**

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it.

Q232 Written

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it.

Classified Practice

Triangles and Lines

Q233 Written

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it.

Q234 Written

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it.

Classified Practice

Triangles and Lines

Q235 Written

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it.

Q236 MCQ

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-3/2, 1/2)$

B $(-1/2, -3/2)$

C $(1/2, 3/2)$

D $(1/2, -3/2)$

Classified Practice

Triangles and Lines

Q237 MCQ

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-2, -3)$

B $(-2, 3)$

C $(3, -2)$

D $(2, 3)$

Q238 MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-3, 3)$

B $(-3, -3)$

C $(3, 3)$

D $(3, -3)$

Classified Practice

Triangles and Lines

Q239 MCQ

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-1, 1)$

B $(1, -1)$

C 0

D $(1, 1)$

Q240 MCQ

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-3, 1)$

B $(1, -3)$

C $(3, 1)$

D $(-3, -1)$

Classified Practice

Triangles and Lines

Q241 **Challenge**

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it.

Classified Practice

Triangles and Lines

Q242 Challenge

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it:

Choose the correct option.

A (4, 1)

B (1, 4)

C (-4, 1)

D (4, -1)

Classified Practice

Triangles and Lines

Q243 Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

Q244 Written

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a .

Classified Practice

Triangles and Lines

Q245 **Written**

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a .

Q246 Written

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a .

Classified Practice

Triangles and Lines

Q247 MCQ

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a:

Choose the correct option.

A $a = -4, -3, \text{ or } 6$

B $a = -3, -3, \text{ or } 6$

C $a = -4, -2, \text{ or } 6$

D $a = -4, -3, \text{ or } 7$

Q248 MCQ

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a:

Choose the correct option.

A $a = -1, -1/2, \text{ or } 1$

B $a = 0, -1/2, \text{ or } 1$

C $a = -1, 1/2, \text{ or } 1$

D $a = -1, -1/2, \text{ or } 2$

Classified Practice

Triangles and Lines

Q249 MCQ

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a :

Choose the correct option.

A $a = 11/10$, $5/2$, or $-9/2$

B $a = 11/10$, $3/2$, or $-7/2$

C $a = 11/10$, $3/2$, or $-9/2$

D $a = 21/10$, $3/2$, or $-9/2$

Q250 MCQ

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a:

Choose the correct option.

A $a = 2, -3/2, \text{ or } 2$

B $a = 1, -1/2, \text{ or } 2$

C $a = 1, -3/2, \text{ or } 3$

D $a = 1, -3/2, \text{ or } 2$

Classified Practice

Triangles and Lines

Q251 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m .

Classified Practice

Triangles and Lines

Q252 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m :

Choose the correct option.

A 15

B 16

C 14

D -15

Classified Practice

Triangles and Lines

Quiz MCQ 1 **Concept Quiz · MCQ**

Determine the shape of triangle A (0, 0), B (3, 0), C (0, 4):

Choose the correct option.

A *isosceles (not right – angled)*

B *scalene (not right – angled)*

C *scalene right – angled*

D *equilateral*

Quiz MCQ 2 Concept Quiz · MCQ

Find the equation of the line through (2,-1) with gradient 3:

Choose the correct option.

A 0

B $y = 3x - 7$

C $y = -3x - 7$

D $y = 3x - 7 + 1$

Classified Practice

Triangles and Lines

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)**Find the perpendicular distance from (1,2) to $3x + 4y - 10 = 0$:****Choose the correct option.**A $6/5$ B $-4/5$ C $-1/5$ D $1/5$

Quiz MCQ 4 Concept Quiz · MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point. Find it:

Choose the correct option.

A $(-3, -3)$

B $(3, 3)$

C $(3, -3)$

D $(-3, 3)$

Classified Practice

Triangles and Lines

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the area of triangle A (0, 0), B (4, 0), C (0, 3).

Quiz WRITTEN 6 Concept Quiz · Written

Find the perpendicular bisector of the segment joining A (1, 2) and B (5, 4).

Classified Practice

Triangles and Lines

Quiz WRITTEN 7 Concept Quiz · Written

Find the equation of the line through (1,2) perpendicular to $y = 3x - 1$.

Quiz WRITTEN 8 Concept Quiz · Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

Answer key

Use the question number shown on each page.

Q001 <i>equilateral</i>	Q027 $x = -3$	Q053 $x = -5$	Q079 C
Q002 <i>equilateral</i>	Q028 $x = 3$	Q054 $y = 3$	Q080 D
Q003 D	Q029 $x = 2$	Q055 $y = -2$	Q081 B
Q004 C	Q030 $x = 4$	Q056 $y = 6$	Q082 D
Q005 <i>isosceles right – angled; right angle at A</i>	Q031 $x = -6$	Q057 $y = 4x - 2$	Q083 $y = 2$
Q006 <i>isosceles right – angled; right angle at A</i>	Q032 B	Q058 $y = 2x - 6$	Q084 D
Q007 B	Q033 C	Q059 B	Q085 $a = 7$
Q008 D	Q034 D	Q060 C	Q086 $a = -1$
Q009 <i>scalene right – angled; right angle at A</i>	Q035 D	Q061 D	Q087 $a = 1$
Q010 D	Q036 B	Q062 A	Q088 $a = 7$
Q011 <i>isosceles right – angled; right angle at B</i>	Q037 D	Q063 B	Q089 $a = 5$
Q012 <i>isosceles right – angled; right angle at B</i>	Q038 B	Q064 D	Q090 D
Q013 <i>isosceles right – angled; right angle at C</i>	Q039 C	Q065 C	Q091 D
Q014 B	Q040 D	Q066 A	Q092 C
Q015 C	Q041 D	Q067 C	Q093 D
Q016 B	Q042 B	Q068 D	Q094 D
Q017 $k = 2$	Q043 D	Q069 D	Q095 $x + y - 2 = 0$
Q018 C	Q044 A	Q070 C	Q096 $x + 3y - 23 = 0$
Q019 $y = -4$	Q045 $x = -3$	Q071 D	Q097 $x + y - 1 = 0$
Q020 $x = -3$	Q046 $y = 3$	Q072 D	Q098 $2x - 5y + 1 = 0$
Q021 $y = 6$	Q047 $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$)	Q073 $y = 3x - 7$	Q099 B
Q022 $x = 3$	Q048 $y = -2$	Q074 $y = 2x + 10$	Q100 A
Q023 $y = 5$	Q049 $x = -4$	Q075 $y = -x + 4$	Q101 B
Q024 $y = 1$	Q050 $7x + 2y + 26 = 0$	Q076 $y = x - 5$	Q102 D
Q025 $y = -7$	Q051 $x = 4$	Q077 $y = -2x + 3$	Q103 $3x + 13y = 0$
Q026 $x = 1$	Q052 $x = -1$	Q078 D	Q104 C

Q105 $m = 2$	Q137 C	Q169 A	Q201 $\sqrt{10}/5$
Q106 $m = -3$	Q138 C	Q170 D	Q202 A
Q107 $m = 1/3$	Q139 D	Q171 $m/k = 2 (k = 3, m = 6)$	Q203 D
Q108 $m = 2$	Q140 A	Q172 D	Q204 D
Q109 $m = -1/3$	Q141 D	Q173 $x + 3y - 18 = 0$	Q205 B
Q110 $m = 3$	Q142 C	Q174 $2x - 3y - 5 = 0$	Q206 D
Q111 $m = 4$	Q143 C	Q175 $2x - y + 2 = 0$	Q207 D
Q112 $m = -4$	Q144 A	Q176 $3x - 2y - 6 = 0$	Q208 A
Q113 $m = 3$	Q145 D	Q177 $2x - y + 1 = 0$	Q209 B
Q114 $m = -4$	Q146 A	Q178 A	Q210 D
Q115 $m = -1/4$	Q147 B	Q179 C	Q211 $\sqrt{10}/2$
Q116 $m = 3$	Q148 D	Q180 C	Q212 $\sqrt{10}$
Q117 $m = -3$	Q149 B	Q181 D	Q213 D
Q118 $m = 1$	Q150 B	Q182 A	Q214 B
Q119 $m = -3$	Q151 $3x - 2y - 14 = 0$	Q183 $B = (2, 1)$	Q215 $\sqrt{10}$
Q120 $m = 1$	Q152 $2x + y + 1 = 0$	Q184 $B = (4, 2)$	Q216 A
Q121 $m = -1/3$	Q153 $2x - y - 4 = 0$	Q185 $B = (3, 6)$	Q217 10
Q122 $m = 2$	Q154 $2x + 3y - 4 = 0$	Q186 $B = (-2, 2)$	Q218 30
Q123 $m = -1$	Q155 $3x + 2y - 7 = 0$	Q187 A	Q219 8
Q124 $m = 3/2$	Q156 C	Q188 A	Q220 9
Q125 $m = -1$	Q157 D	Q189 C	Q221 34
Q126 $m = -2/3$	Q158 C	Q190 C	Q222 7
Q127 $m = -1/2$	Q159 D	Q191 $B = (-3, 4)$	Q223 D
Q128 D	Q160 D	Q192 C	Q224 C
Q129 C	Q161 $5x - 3y + 25 = 0$	Q193 $3/\sqrt{13} = 3\sqrt{13}/13$	Q225 A
Q130 A	Q162 $3x - 2y - 4 = 0$	Q194 $\sqrt{5}$	Q226 B
Q131 B	Q163 $x - 2y - 12 = 0$	Q195 $6/\sqrt{13} = 6\sqrt{13}/13$	Q227 B
Q132 A	Q164 $2x + 5y - 13 = 0$	Q196 6	Q228 D
Q133 A	Q165 $2x + 3y + 19 = 0$	Q197 $4\sqrt{5}$	Q229 5
Q134 B	Q166 D	Q198 1	Q230 D
Q135 D	Q167 D	Q199 $\sqrt{13}$	Q231 $(1/2, -3/2)$
Q136 A	Q168 B	Q200 10	Q232 $(-2, 3)$

Q233 (3, -3)

Q234 (1, 1)

Q235 (-3, 1)

Q236 D

Q237 B

Q238 D

Q239 D

Q240 A

Q241 (4, 1)

Q242 A

Q243 $a = -4, -3, \text{ or } 6$ Q244 $a = -1, -1/2, \text{ or } 1$ Q245 $a = 11/10, 3/2, \text{ or } -9/2$ Q246 $a = 1, -3/2, \text{ or } 2$

Q247 A

Q248 A

Q249 C

Q250 D

Q251 15

Q252 A

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 D

Quiz MCQ 4 C

Quiz WRITTEN 5 6

Quiz WRITTEN 6 $y = -2x + 9$ Quiz WRITTEN 7 $y = -x/3 + 7/3$ Quiz WRITTEN 8 $a = -4, -3, \text{ or } 6$

Concept 3 | Circles

English Student Edition • Focused formulas and guided practice

Formula opener

Circle

$$(x - a)^2 + (y - b)^2 = r^2$$

General form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Tangent

$$xx_1 + yy_1 = r^2$$

Classified Practice

Circles

Q001 **Written****Find the equation of the circle with centre (3,-2) and radius 5.**

Q002 Written**Find the equation of the circle with centre $(-3,1)$ and radius 2.**

Classified Practice

Circles

Q003 **Written****Find the equation of the circle with centre the origin and radius 4.**

Q004 Written**Find the equation of the circle with centre (3,4), radius 6.**

Classified Practice

Circles

Q005 Written

Find the equation of the circle with centre $(-2,6)$, radius $2\sqrt{3}$.

Q006 Written**Find the equation of the circle with centre the origin, radius 5.**

Classified Practice

Circles

Q007 MCQ

Find the equation of the circle with centre (3,-2) and radius 5:**Choose the correct option.**

A $(x - 3)^2 + (y + 2)^2 = 25$

B $(x + 3)^2 + (y + 2)^2 = 25$

C $(x - 3)^2 + (y + 2)^2 = 25 + 1$

D 4

Q008 MCQ**Find the equation of the circle with centre (-3,1) and radius 2:****Choose the correct option.**

A $(x + 3)^2 + (y - 1)^2 = 4$

B $(x + 3)^2 + (y + 1)^2 = 4$

C $(x + 3)^2 + (y - 1)^2 = 4 + 1$

D 6

Classified Practice

Circles

Q009 MCQ

Find the equation of the circle with centre the origin and radius 4:**Choose the correct option.**

A $x^2 + y^2 = 16 + 1$

B 6

C 7

D $x^2 + y^2 = 16$

Q010 MCQ**Find the equation of the circle with centre (3,4), radius 6:****Choose the correct option.****A** 9

B $(x - 3)^2 + (y - 4)^2 = 36$

C $(x + 3)^2 + (y - 4)^2 = 36$

D $(x - 3)^2 + (y - 4)^2 = 36 + 1$

Classified Practice

Circles

Q011 MCQ

Find the equation of the circle with centre $(-2,6)$, radius $2\sqrt{3}$:**Choose the correct option.**

A 10

B $(x + 2)^2 + (y - 6)^2 = 12$

C $(x + 2)^2 + (y + 6)^2 = 12$

D $(x + 2)^2 + (y - 6)^2 = 12 + 1$

Q012 MCQ**Find the equation of the circle with centre the origin, radius 5:****Choose the correct option.**

A $x^2 + y^2 = 25$

B $x^2 + y^2 = 25 + 1$

C 10

D 11

Classified Practice

Circles

Q013 Written

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents.

Q014 Written**Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents.**

Classified Practice

Circles

Q015 **Written**

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents.

Q016 Written**Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents.**

Classified Practice

Circles

Q017 Written

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents.

Q018 Written**Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents.**

Classified Practice

Circles

Q019 MCQ

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents:**Choose the correct option.**A circle; centre $(3, -1)$, radius 2

B 3

C 4

D 5

Classified Practice

Circles

Q020 MCQ

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents:**Choose the correct option.**

A 7

B 8

C circle; centre (2, 3), radius 3

D 6

Classified Practice

Circles

Q021 MCQ

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents:**Choose the correct option.**

- A 10
- B 11
- C 12
- D circle; centre $(2, -4)$, radius 4

Classified Practice

Circles

Q022 MCQ

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents:**Choose the correct option.**

A 12

B 13

C circle; centre (2, 0), radius 2

D 11

Classified Practice

Circles

Q023 MCQ

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents:**Choose the correct option.**

- A 13
- B 14
- C circle; centre $(-1, 2)$, radius 5
- D circle; centre $(-1, 2)$, radius 10

Classified Practice

Circles

Q024 MCQ

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents:**Choose the correct option.**

A 14

B 15

C circle; centre $(-2, 3)$, radius $\sqrt{13}$

D 13

Classified Practice

Circles

Q025 **Challenge**

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius.

Classified Practice

Circles

Q026 Challenge

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius:

Choose the correct option.

A 0

B centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$

C centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13 + 1$

D centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13 - 1$

Classified Practice

Circles

Q027 **Written****Find the equation of the circle with centre (2,3) passing through (4,6).**

Q028 Written**Find the equation of the circle with centre $(-3,1)$ passing through $(4,-2)$.**

Classified Practice

Circles

Q029 Written

Find the equation of the circle with centre $(1,-3)$ through $(-3,0)$.

Q030 Written**Find the equation of the circle with centre $(-4,1)$ through $(3,-2)$.**

Classified Practice

Circles

Q031 **Written****Find the equation of the circle with centre (2,-1) through the origin.**

Q032 Written**Find the equation of the circle with centre the origin through (4,-3).**

Classified Practice

Circles

Q033 MCQ

Find the equation of the circle with centre (2,3) passing through (4,6):**Choose the correct option.**

A $(x - 2)^2 + (y - 3)^2 = 13 + 1$

B 4

C $(x - 2)^2 + (y - 3)^2 = 13$

D $(x + 2)^2 + (y - 3)^2 = 13$

Q034 MCQ**Find the equation of the circle with centre $(-3,1)$ passing through $(4,-2)$:****Choose the correct option.**

A $(x + 3)^2 + (y - 1)^2 = 58$

B $(x + 3)^2 + (y + 1)^2 = 58$

C $(x + 3)^2 + (y - 1)^2 = 58 + 1$

D 7

Classified Practice

Circles

Q035 MCQ

Find the equation of the circle with centre (1,-3) through (-3,0):**Choose the correct option.**

A $(x + 1)^2 + (y + 3)^2 = 25$

B $(x - 1)^2 + (y + 3)^2 = 25 + 1$

C 10

D $(x - 1)^2 + (y + 3)^2 = 25$

Q036 MCQ**Find the equation of the circle with centre $(-4,1)$ through $(3,-2)$:****Choose the correct option.**

A $(x + 4)^2 + (y - 1)^2 = 58$

B $(x + 4)^2 + (y + 1)^2 = 58$

C $(x + 4)^2 + (y - 1)^2 = 58 + 1$

D 11

Classified Practice

Circles

Q037 MCQ

Find the equation of the circle with centre (2,-1) through the origin:**Choose the correct option.**

A 12

B $(x - 2)^2 + (y + 1)^2 = 5$ C $(x + 2)^2 + (y + 1)^2 = 5$ D $(x - 2)^2 + (y + 1)^2 = 5 + 1$

Q038 MCQ**Find the equation of the circle with centre the origin through (4,-3):****Choose the correct option.**

A $x^2 + y^2 = 25$

B $x^2 + y^2 = 25 + 1$

C 12

D 13

Classified Practice

Circles

Q039 **Written****Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5).**

Q040 Written**Find the equation of the circle with diameter endpoints $(3,4)$, $(5,-2)$.**

Classified Practice

Circles

Q041 **Written****Find the equation of the circle with diameter endpoints $(4,7)$, $(-2,-1)$.**

Q042 Written**Find the equation of the circle with diameter endpoints $(4,-2)$, $(-6,-2)$.**

Classified Practice

Circles

Q043 **Written****Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$.**

Q044 Written**Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$.**

Classified Practice

Circles

Q045 MCQ

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5):**Choose the correct option.**

A $(x - 2)^2 + (y + 1)^2 = 17$

B $(x + 2)^2 + (y + 1)^2 = 17$

C $(x - 2)^2 + (y + 1)^2 = 17 + 1$

D 5

Q046 MCQ**Find the equation of the circle with diameter endpoints (3,4), (5,-2):****Choose the correct option.**

A $(x - 4)^2 + (y - 1)^2 = 10 + 1$

B 8

C $(x - 4)^2 + (y - 1)^2 = 10$

D $(x + 4)^2 + (y - 1)^2 = 10$

Classified Practice

Circles

Q047 MCQ

Find the equation of the circle with diameter endpoints (4,7), (-2,-1):**Choose the correct option.**

A $(x - 1)^2 + (y - 3)^2 = 25 + 1$

B 14

C $(x - 1)^2 + (y - 3)^2 = 25$

D $(x + 1)^2 + (y - 3)^2 = 25$

Q048 MCQ**Find the equation of the circle with diameter endpoints (4,-2), (-6,-2):****Choose the correct option.**

A $(x + 1)^2 + (y + 2)^2 = 25$

B $(x + 1)^2 + (y + 2)^2 = 25 + 1$

C 14**D** 15

Classified Practice

Circles

Q049 MCQ

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$:**Choose the correct option.**

A 16

B $(x + 2)^2 + (y - 1)^2 = 2$ C $(x + 2)^2 + (y + 1)^2 = 2$ D $(x + 2)^2 + (y - 1)^2 = 2 + 1$

Q050 MCQ**Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$:****Choose the correct option.**

A $x^2 + (y - 2)^2 = 25 + 1$

B 17

C $x^2 + (y - 2)^2 = 25$

D $x^2 + (y + 2)^2 = 25$

Classified Practice

Circles

Q051 **Written****Find the equation of the circle passing through (1,1), (2,-1), and (3,2).**

Q052 Written**Find the equation of the circle passing through (0,1), (-1,2), and (3,1).**

Classified Practice

Circles

Q053 Written

Find the equation of the circle with through $(0,0)$, $(-1,1)$, $(2,4)$.

Q054 Written**Find the equation of the circle with through $(-4,6)$, $(0,8)$, $(3,-1)$.**

Classified Practice

Circles

Q055 **Written****Find the equation of the circle with through $(1,1)$, $(5,-1)$, $(-3,-7)$.**

Q056 MCQ**Find the equation of the circle passing through (1,1), (2,-1), and (3,2):****Choose the correct option.**

A $x^2 + y^2 - 5x - y + 4 = 0$

B $x^2 + y^2 + 5x - y + 4 = 0$

C $x^2 + y^2 - 5x - y + 4 = 0 + 1$

D 6

Classified Practice

Circles

Q057 MCQ

Find the equation of the circle passing through (0,1), (-1,2), and (3,1):**Choose the correct option.**

A 9

B $x^2 + y^2 - 3x - 7y + 6 = 0$

C $x^2 + y^2 + 3x - 7y + 6 = 0$

D $x^2 + y^2 - 3x - 7y + 6 = 0 + 1$

Q058 MCQ**Find the equation of the circle with through (0,0), (-1,1), (2,4):****Choose the correct option.**

A $x^2 + y^2 + 2x - 4y = 0$

B $x^2 + y^2 - 2x - 4y = 0 + 1$

C 18

D $x^2 + y^2 - 2x - 4y = 0$

Classified Practice

Circles

Q059 MCQ

Find the equation of the circle with through $(-4,6)$, $(0,8)$, $(3,-1)$:**Choose the correct option.**

A $x^2 + y^2 + 6y - 16 = 0$

B $x^2 + y^2 - 6y - 16 = 0 + 1$

C 19

D $x^2 + y^2 - 6y - 16 = 0$

Q060 MCQ**Find the equation of the circle with through (1,1), (5,-1), (-3,-7):****Choose the correct option.**

A $x^2 + y^2 + 2x + 8y - 8 = 0$

B $x^2 + y^2 - 2x + 8y - 8 = 0 + 1$

C 20

D $x^2 + y^2 - 2x + 8y - 8 = 0$

Classified Practice

Circles

Q061 **Challenge**

Find the circle through $(0,2)$, $(2,0)$, and $(4,6)$, giving its centre and radius.

Classified Practice

Circles

Q062 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius:

Choose the correct option.

A $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$

B $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10} + 1$

C $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10} - 1$

D 0

Classified Practice

Circles

Q063 Written

Find the circle with centre $(4, -2)$ tangent to the x-axis.

Q064 Written**Find the circle with centre $(-2,3)$ tangent to the y -axis.**

Classified Practice

Circles

Q065 Written

Find the circle with centre $(-1,4)$ tangent to x-axis.

Q066 Written**Find the circle with centre (3,-4) tangent to x-axis.**

Classified Practice

Circles

Q067 **Written****Find the circle with centre (5,-4) tangent to x-axis.**

Q068 Written**Find the circle with centre $(-1,6)$ tangent to y -axis.**

Classified Practice

Circles

Q069 **Written****Find the circle with centre $(2, -5)$ tangent to y -axis.**

Q070 Written**Find the circle with centre $(-3, -4)$ tangent to y -axis.**

Classified Practice

Circles

Q071 MCQ

Find the circle with centre (4,-2) tangent to the x-axis:**Choose the correct option.**

A radius zero

B $(x - 4)^2 + (y + 2)^2 = 4$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q072 MCQ

Find the circle with centre $(-2,3)$ tangent to the y -axis:**Choose the correct option.**

- A radius zero
- B $(x + 2)^2 + (y - 3)^2 = 4$
- C no real circle
- D centre on the wrong locus

Classified Practice

Circles

Q073 MCQ

Find the circle with centre $(-1,4)$ tangent to x-axis:**Choose the correct option.**

- A radius zero
- B $(x + 1)^2 + (y - 4)^2 = 16$
- C no real circle
- D centre on the wrong locus

Classified Practice

Circles

Q074 MCQ

Find the circle with centre (3,-4) tangent to x-axis:**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 3)^2 + (y + 4)^2 = 16$

Classified Practice

Circles

Q075 MCQ

Find the circle with centre (5,-4) tangent to x-axis:**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 5)^2 + (y + 4)^2 = 16$

Classified Practice

Circles

Q076 MCQ

Find the circle with centre $(-1,6)$ tangent to y-axis:**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x + 1)^2 + (y - 6)^2 = 1$

Classified Practice

Circles

Q077 MCQ

Find the circle with centre (2,-5) tangent to y-axis:**Choose the correct option.**

A radius zero

B $(x - 2)^2 + (y + 5)^2 = 4$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q078 MCQ

Find the circle with centre $(-3,-4)$ tangent to y-axis:**Choose the correct option.**

- A centre on the wrong locus
- B radius zero
- C $(x + 3)^2 + (y + 4)^2 = 9$
- D no real circle

Classified Practice

Circles

Q079 **Written****Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis.**

Q080 Written**Find the circles through (1,2) tangent to both axes.**

Classified Practice

Circles

Q081 **Written****Find the circle with through $(-1, -2)$ tangent to both axes.**

Q082 Written**Find the circle with through $(-2,-1)$ tangent to both axes.**

Classified Practice

Circles

Q083 Written

Find the circle with through $(8,-1)$ tangent to both axes.

Classified Practice

Circles

Q084 MCQ

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis:**Choose the correct option.**

A centre on the wrong locus

B radius zero

C $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$

D no real circle

Classified Practice

Circles

Q085 MCQ

Find the circles through (1,2) tangent to both axes:**Choose the correct option.**

A no real circle

B centre on the wrong locus

C radius zero

D $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$

Classified Practice

Circles

Q086 MCQ

Find the circle with through $(-1,-2)$ tangent to both axes:**Choose the correct option.**

A centre on the wrong locus

B radius zero

C $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$

D no real circle

Classified Practice

Circles

Q087 MCQ

Find the circle with through $(-2,-1)$ tangent to both axes:**Choose the correct option.**

A centre on the wrong locus

B radius zero

C $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$

D no real circle

Classified Practice

Circles

Q088 MCQ

Find the circle with through (8,-1) tangent to both axes:**Choose the correct option.**

A radius zero

B $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q089 **Written**

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3).

Q090 Written

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2).

Classified Practice

Circles

Q091 **Written****Find the circle with centre on $y = x + 1$ through (3,4) and (1,4).**

Q092 Written**Find the circle with centre on $y = 2x - 6$ through $(3,7)$ and $(-1,3)$.**

Classified Practice

Circles

Q093 MCQ

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3):**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 1)^2 + (y + 1)^2 = 25$

Classified Practice

Circles

Q094 MCQ

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2):**Choose the correct option.**

- A centre on the wrong locus
- B radius zero
- C $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$
- D no real circle

Classified Practice

Circles

Q095 MCQ

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4):**Choose the correct option.**

- A centre on the wrong locus
- B radius zero
- C $(x - 2)^2 + (y - 3)^2 = 2$
- D no real circle

Classified Practice

Circles

Q096 MCQ

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3):**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 4)^2 + (y - 2)^2 = 26$

Classified Practice

Circles

Q097 **Challenge**

A circle centre is constrained to the x-axis and passes through $P1(-2,2)$, $P2(1,1)$. Find its standard equation, centre and radius.

Classified Practice

Circles

Q098 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius:

Choose the correct option.

A $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5} - 1$

B 0

C $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$

D $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5} + 1$

Classified Practice

Circles

Q099 **Written**

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle.

Q100 Written

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle.

Classified Practice

Circles

Q101 **Written**

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle.

Q102 Written

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle.

Classified Practice

Circles

Q103 Written

Find the range of k for $x^2 + y^2 + 2kx - 2(k+3)y + 4k^2 + 1 = 0$ to represent a circle.

Classified Practice

Circles

Q104 MCQ

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle:

Choose the correct option.

- A all real parameter values
- B radius squared is negative
- C $k < -1$ or $k > 3$
- D radius squared is zero

Classified Practice

Circles

Q105 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle:

Choose the correct option.

- A radius squared is negative
- B $k < -\sqrt{6}$ or $k > \sqrt{6}$
- C radius squared is zero
- D all real parameter values

Classified Practice

Circles

Q106 MCQ

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle:**Choose the correct option.**

- A all real parameter values
- B radius squared is negative
- C $k < 5/4$
- D radius squared is zero

Classified Practice

Circles

Q107 MCQ

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle:

Choose the correct option.

- A radius squared is negative
- B $k < -1/5$ or $k > 3$
- C radius squared is zero
- D all real parameter values

Classified Practice

Circles

Q108 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 2(k + 3)y + 4k^2 + 1 = 0$ to represent a circle:

Choose the correct option.

- A radius squared is zero
- B all real parameter values
- C radius squared is negative
- D $-1 < k < 4$

Classified Practice

Circles

Q109 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k-range giving a positive radius.

Classified Practice

Circles

Q110 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k -range giving a positive radius:

Choose the correct option.

A 0

B $0 < k < 4$

C $0 < k < 4 + 1$

D $0 < k < 4 - 1$

Classified Practice

Circles

Q111 Written

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$.

Q112 Written

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$.

Classified Practice

Circles

Q113 Written

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$.

Q114 Written

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$.

Classified Practice

Circles

Q115 Written

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$.

Classified Practice

Circles

Q116 MCQ

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$:

Choose the correct option.

- A no real intersection
- B two intersection points
- C $(1, -1)$ and $(-1, 3)$
- D tangent

Classified Practice

Circles

Q117 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$:**Choose the correct option.**

- A tangent
- B no real intersection
- C two intersection points
- D (0, 2) and (2, 0)

Classified Practice

Circles

Q118 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$:**Choose the correct option.**

- A no real intersection
- B two intersection points
- C $(3, -1)$
- D tangent

Classified Practice

Circles

Q119 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D $(-2, 1)$

Classified Practice

Circles

Q120 MCQ

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$:

Choose the correct option.

- A two intersection points
- B $(-2, 1)$ and $(4, -1)$
- C tangent
- D no real intersection

Classified Practice

Circles

Q121 Written

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$.

Q122 Written

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$.

Classified Practice

Circles

Q123 Written

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$.

Q124 Written**Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$.**

Classified Practice

Circles

Q125 Written

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$.

Classified Practice

Circles

Q126 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$:**Choose the correct option.**

- A tangent
- B no real intersection
- C two intersection points
- D 0points

Classified Practice

Circles

Q127 MCQ

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$:**Choose the correct option.**

A 2points : (1, -3) and (3, 1)

B tangent

C no real intersection

D two intersection points

Classified Practice

Circles

Q128 MCQ

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$:**Choose the correct option.**

A no real intersection

B two intersection points

C 2 points : $\left(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2\right)$ and $\left(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2\right)$

D tangent

Classified Practice

Circles

Q129 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$:**Choose the correct option.**

- A no real intersection
- B two intersection points
- C 0points
- D tangent

Classified Practice

Circles

Q130 MCQ

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D 1 point : $(-2, 0)$

Classified Practice

Circles

Q131 Written

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k .

Q132 Written

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r .

Classified Practice

Circles

Q133 Written

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m .

Q134 Written

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range.

Classified Practice

Circles

Q135 Written

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m .

Q136 Written

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n .

Classified Practice

Circles

Q137 **Written**

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m .

Q138 Written

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r .

Classified Practice

Circles

Q139 Written

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range.

Classified Practice

Circles

Q140 MCQ

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k :

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D $-5 < k < 5$

Classified Practice

Circles

Q141 MCQ

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r :

Choose the correct option.

A two intersection points

B $r = \sqrt{5}$

C tangent

D no real intersection

Classified Practice

Circles

Q142 MCQ

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m :

Choose the correct option.

A two intersection points

B $m = \pm\sqrt{3}$

C tangent

D no real intersection

Classified Practice

Circles

Q143 MCQ

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range:

Choose the correct option.

A two intersection points

B $0 < r < \sqrt{10}$

C tangent

D no real intersection

Classified Practice

Circles

Q144 MCQ

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m :

Choose the correct option.

- A no real intersection
- B two intersection points
- C $m = \pm 4$
- D tangent

Classified Practice

Circles

Q145 MCQ

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n :

Choose the correct option.

A $n < -10$ or $n > 10$

B tangent

C no real intersection

D two intersection points

Classified Practice

Circles

Q146 MCQ

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m :

Choose the correct option.

- A no real intersection
- B two intersection points
- C $-5 \leq m \leq 5$
- D tangent

Classified Practice

Circles

Q147 MCQ

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r :

Choose the correct option.

A two intersection points

B $r = \sqrt{2}$

C tangent

D no real intersection

Classified Practice

Circles

Q148 MCQ

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D $r > \sqrt{17}$

Classified Practice

Circles

Q149 **Challenge**

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r .

Classified Practice

Circles

Q150 Challenge

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r :

Choose the correct option.

A $0 < r < 2 + 1$

B $0 < r < 2 - 1$

C 0

D $0 < r < 2$

Classified Practice

Circles

Q151 **Written****Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$.**

Q152 Written**Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$.**

Classified Practice

Circles

Q153 **Written****Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$.**

Q154 Written**Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$.**

Classified Practice

Circles

Q155 **Written****Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$.**

Q156 MCQ

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$:

Choose the correct option.

A $\sqrt{14}$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

Classified Practice

Circles

Q157 MCQ

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$:

Choose the correct option.

A $\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 - d^2} + 1$

C 4

D $2\sqrt{r^2 + d^2}$

Q158 MCQ

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$:

Choose the correct option.

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D $2\sqrt{5}$

Classified Practice

Circles

Q159 MCQ

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$:

Choose the correct option.

A $2\sqrt{r^2 - d^2} + 1$

B $3\sqrt{2}$

C $2\sqrt{r^2 + d^2}$

D $\sqrt{r^2 - d^2}$

Q160 MCQ**Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$:****Choose the correct option.**

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D 2

Classified Practice

Circles

Q161 **Written**

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2.

Q162 Written

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$.

Classified Practice

Circles

Q163 **Written**

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$.

Q164 Written

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$.

Classified Practice

Circles

Q165 MCQ

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2:

Choose the correct option.

A $\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 - d^2} + 1$

C $k = \pm 2$

D $2\sqrt{r^2 + d^2}$

Q166 MCQ

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$:

Choose the correct option.

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D $k = \pm\sqrt{31}$

Classified Practice

Circles

Q167 MCQ

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$:

Choose the correct option.

A $k = \pm 2$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

Classified Practice

Circles

Q168 MCQ

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$:

Choose the correct option.

A $2\sqrt{r^2 - d^2} + 1$

B $k = 0$ or $k = -2$

C $2\sqrt{r^2 + d^2}$

D $\sqrt{r^2 - d^2}$

Classified Practice

Circles

Q169 **Challenge**

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4.

Classified Practice

Circles

Q170 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4:

Choose the correct option.

A $k = -1$ or $k = 7$

B $k = -1$ or $k = 7 + 1$

C $k = -1$ or $k = 7 - 1$

D 0

Classified Practice

Circles

Q171 **Written****Find the tangent to $x^2 + y^2 = 25$ at (4,3).**

Q172 Written**Find the tangent to $x^2 + y^2 = 34$ at (5,3).**

Classified Practice

Circles

Q173 **Written****Find the tangents $x^2 + y^2 = 10$ at (3,-1).**

Q174 Written**Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$.**

Classified Practice

Circles

Q175 Written

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$.

Classified Practice

Circles

Q176 MCQ

Find the tangent to $x^2 + y^2 = 25$ at (4,3):**Choose the correct option.**

A $4x + 3y = 25$

B $4x - 3y = 25$

C $4x + 3y = 25 + 1$

D the radius line

Classified Practice

Circles

Q177 MCQ

Find the tangent to $x^2 + y^2 = 34$ at (5,3):**Choose the correct option.**

- A $5x + 3y = 34 + 1$
- B the radius line
- C $5x + 3y = 34$
- D $5x - 3y = 34$

Classified Practice

Circles

Q178 MCQ

Find the tangents $x^2 + y^2 = 10$ at (3,-1):**Choose the correct option.**

A $3x - y = 10$

B $3x - y = 10 + 1$

C the radius line

D 8

Classified Practice

Circles

Q179 MCQ

Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$:**Choose the correct option.**

A the radius line

B 9

C $-5x - 2y = 29$ D $-5x - 2y = 29 + 1$

Classified Practice

Circles

Q180 MCQ

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$:**Choose the correct option.**

A $-2x + y = 5$

B $-2x - y = 5$

C $-2x + y = 5 + 1$

D the radius line

Classified Practice

Circles

Q181 **Written****Find the tangents from (1,3) to $x^2 + y^2 = 5$.**

Q182 Written**Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$.**

Classified Practice

Circles

Q183 Written

Find the tangents from (-1,3) to $x^2 + y^2 = 5$.

Q184 Written**Find the tangents from (2,1) to $x^2 + y^2 = 1$.**

Classified Practice

Circles

Q185 **Written****Find the tangents from (2,6) to $x^2 + y^2 = 20$.**

Classified Practice

Circles

Q186 MCQ

Find the tangents from (1,3) to $x^2 + y^2 = 5$:**Choose the correct option.**

- A $2x - y = 5$ and $-x + 2y = 5$
 - B $2x + y = 5$ and $-x + 2y = 5 + 1$
 - C the radius line
 - D $2x + y = 5$ and $-x + 2y = 5$
-
-
-

Classified Practice

Circles

Q187 MCQ

Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$:**Choose the correct option.**

- A $-3x + y = 10$ and $x + 3y = 10$
- B $-3x - y = 10$ and $x + 3y = 10$
- C $-3x + y = 10$ and $x + 3y = 10 + 1$
- D the radius line

Classified Practice

Circles

Q188 MCQ

Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$:**Choose the correct option.**

- A $-2x - y = 5$ and $x + 2y = 5$
- B $-2x + y = 5$ and $x + 2y = 5 + 1$
- C the radius line
- D $-2x + y = 5$ and $x + 2y = 5$

Classified Practice

Circles

Q189 MCQ

Find the tangents from (2,1) to $x^2 + y^2 = 1$:**Choose the correct option.**

A 12

B $y = 1$ and $4x - 3y = 5$ C $y = 1$ and $4x - 3y = 5 + 1$

D the radius line

Classified Practice

Circles

Q190 MCQ

Find the tangents from (2,6) to $x^2 + y^2 = 20$:**Choose the correct option.**

A $-x + 2y = 10$ and $2x + y = 10 + 1$

B the radius line

C $-x + 2y = 10$ and $2x + y = 10$

D $-x - 2y = 10$ and $2x + y = 10$

Classified Practice

Circles

Q191 **Challenge**

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points.

Classified Practice

Circles

Q192 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points:

Choose the correct option.

A 0

B $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

C $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2) + 1$

D $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2) - 1$

Classified Practice

Circles

Q193 **Written****Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$.**

Q194 Written**Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E.**

Classified Practice

Circles

Q195 **Written****Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E.**

Q196 Written**Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E.**

Classified Practice

Circles

Q197 **Written****Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E.**

Q198 Written**Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E.**

Classified Practice

Circles

Q199 Written

Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E.

Classified Practice

Circles

Q200 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$., the required value is:

Choose the correct option.

- A internally tangent
- B intersecting at two points
- C intersect at 2 points
- D separate

Classified Practice

Circles

Q201 MCQ

For the expression Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E., the required value is:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D touch externally

Classified Practice

Circles

Q202 MCQ

For the expression Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E., the required value is:

Choose the correct option.

- A intersecting at two points
- B E : intersect at 2 points
- C separate
- D internally tangent

Classified Practice

Circles

Q203 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E., the required value is:

Choose the correct option.

- A C : touch internally
- B separate
- C internally tangent
- D intersecting at two points

Classified Practice

Circles

Q204 MCQ

For the expression Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E., the required value is:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D touch internally

Classified Practice

Circles

Q205 MCQ

For the expression Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E., the required value is:

Choose the correct option.

- A intersecting at two points
- B E : intersect at 2 points
- C separate
- D internally tangent

Classified Practice

Circles

Q206 MCQ

For the expression Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E., the required value is:

Choose the correct option.

- A internally tangent
- B intersecting at two points
- C D : one is inside the other
- D separate

Classified Practice

Circles

Q207 **Written**

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally.

Q208 Written**Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally.**

Classified Practice

Circles

Q209 Written

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally.

Classified Practice

Circles

Q210 MCQ

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally:**Choose the correct option.**

A $(x - 3)^2 + (y - 1)^2 = 45$

B separate

C internally tangent

D intersecting at two points

Classified Practice

Circles

Q211 MCQ

Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D $(x + 3)^2 + (y + 9)^2 = 16$

Classified Practice

Circles

Q212 MCQ

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally:**Choose the correct option.**

A $(x - 3)^2 + (y + 9)^2 = 144$

B separate

C internally tangent

D intersecting at two points

Classified Practice

Circles

Q213 **Challenge**

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally.

Classified Practice

Circles

Q214 Challenge

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally:

Choose the correct option.

A $(x + 1)^2 + (y + 2)^2 = 9 + 1$

B $(x + 1)^2 + (y + 2)^2 = 9 - 1$

C 0

D $(x + 1)^2 + (y + 2)^2 = 9$

Classified Practice

Circles

Q215 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$.

Q216 Written

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$.

Classified Practice

Circles

Q217 Written

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$.

Q218 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$.

Classified Practice

Circles

Q219 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$:

Choose the correct option.

A $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)

B $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4 + 1$)

C the radical axis only

D 4

Classified Practice

Circles

Q220 MCQ

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$:

Choose the correct option.

- A $x + y - 1 = 0$
- B the radical axis only
- C $x + y - 1 = 0$
- D $x - y - 1 = 0$

Classified Practice

Circles

Q221 MCQ

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$:

Choose the correct option.

A 8

B $12x - 6y - 7 = 0$

C $12x - 6y - 7 = 0 + 1$

D the radical axis only

Classified Practice

Circles

Q222 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$:

Choose the correct option.

A $x = 1/2 + 1$

B the radical axis only

C 9

D $x = 1/2$

Classified Practice

Circles

Q223 Written

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$.

Q224 Written**Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$.**

Classified Practice

Circles

Q225 Written

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$.

Classified Practice

Circles

Q226 MCQ

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$:**Choose the correct option.**

A 5

B $(-3, -1)$ and $(-1, 3)$ C $(-3, -1)$ and $(-1, 3) + 1$

D the radical axis only

Classified Practice

Circles

Q227 MCQ

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$:

Choose the correct option.

A the radical axis only

B 7

C (0, 2) and $(8/5, -6/5)$

D (0, 2) and $(8/5, -6/5) + 1$

Classified Practice

Circles

Q228 MCQ

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$:**Choose the correct option.**

A 10

B (1, 2) and (2, 1)

C (1, 2) and (2, 1) + 1

D the radical axis only

Classified Practice

Circles

Q229 **Challenge**

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points.

Classified Practice

Circles

Q230 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points:

Choose the correct option.

A $(-4, 3)$ and $(3, -4) - 1$

B 0

C $(-4, 3)$ and $(3, -4)$

D $(-4, 3)$ and $(3, -4) + 1$

Classified Practice

Circles

Q231 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0).

Q232 Written

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin.

Classified Practice

Circles

Q233 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0).

Q234 Written

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3, 2)$.

Classified Practice

Circles

Q235 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0).

Classified Practice

Circles

Q236 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0):

Choose the correct option.

A $x^2 - y^2 + 2x + 6y - 15 = 0$

B $x^2 + y^2 + 2x + 6y - 15 = 0 + 1$

C the radical axis only

D $x^2 + y^2 + 2x + 6y - 15 = 0$

Classified Practice

Circles

Q237 MCQ

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin:

Choose the correct option.

A $x^2 + y^2 + 4x + 2y = 0$

B $x^2 - y^2 + 4x + 2y = 0$

C $x^2 + y^2 + 4x + 2y = 0 + 1$

D the radical axis only

Classified Practice

Circles

Q238 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0):

Choose the correct option.

- A $x^2 - y^2 + 10x + 20y - 39 = 0$
- B $x^2 + y^2 + 10x + 20y - 39 = 0 + 1$
- C the radical axis only
- D $x^2 + y^2 + 10x + 20y - 39 = 0$

Classified Practice

Circles

Q239 MCQ

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3,2)$:

Choose the correct option.

A $x^2 + y^2 + 24x + 32y - 5 = 0 + 1$

B the radical axis only

C $x^2 + y^2 + 24x + 32y - 5 = 0$

D $x^2 - y^2 + 24x + 32y - 5 = 0$

Classified Practice

Circles

Q240 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0):

Choose the correct option.

A the radical axis only

B $x^2 + y^2 - 10x - 6y + 24 = 0$

C $x^2 - y^2 - 10x - 6y + 24 = 0$

D $x^2 + y^2 - 10x - 6y + 24 = 0 + 1$

Classified Practice

Circles

Q241 **Challenge**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2).

Classified Practice

Circles

Q242 Challenge

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2):

Choose the correct option.

- A $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$
- B $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0 + 1$
- C $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0 - 1$
- D 0

Classified Practice

Circles

Quiz MCQ 1 Concept Quiz · MCQ

For the circle $(x - 2)^2 + (y + 3)^2 = 25$, determine its centre and radius:

Choose the correct option.

- A centre $(-2, -3)$, radius 5
- B centre $(2, -3)$, radius 5
- C centre $(2, 3)$, radius 5
- D centre $(2, -3)$, radius 10

Classified Practice

Circles

Quiz MCQ 2 **Concept Quiz · MCQ**

Determine the substitution discriminant when a line meets a circle at exactly one real point:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D 0

Classified Practice

Circles

Quiz MCQ 3 **Concept Quiz · MCQ**

Determine the chord length for radius r and centre-to-chord distance d :

Choose the correct option.

A $2\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

Classified Practice

Circles

Quiz MCQ 4 **Concept Quiz · MCQ**

Determine the relative position of two non-overlapping circles when $D = r_1 + r_2$:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D externally tangent

Classified Practice

Circles

Quiz WRITTEN 5 Concept Quiz · Written

Find the circle with centre (1,-2) passing through (4,2).

Quiz WRITTEN 6 Concept Quiz · Written

Find the intersections of $x^2 + y^2 = 25$ and $y = 3$.

Classified Practice

Circles

Quiz WRITTEN 7 Concept Quiz · Written

Find the tangent to $x^2 + y^2 = 25$ at (3,4).

Quiz WRITTEN 8 Concept Quiz · Written

The circles $S_1 = x^2 + y^2 - 5 = 0$ and $S_2 = x^2 + y^2 - 2x - 2y - 4 = 0$ meet. Find the circle $S_1 + \lambda S_2 = 0$ passing through (1,1).

Answer key

Use the question number shown on each page.

Q001 $(x - 3)^2 + (y + 2)^2 = 25$	Q027 $(x - 2)^2 + (y - 3)^2 = 13$	Q053 $x^2 + y^2 - 2x - 4y = 0$	Q079 $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$
Q002 $(x + 3)^2 + (y - 1)^2 = 4$	Q028 $(x + 3)^2 + (y - 1)^2 = 58$	Q054 $x^2 + y^2 - 6y - 16 = 0$	Q080 $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$
Q003 $x^2 + y^2 = 16$	Q029 $(x - 1)^2 + (y + 3)^2 = 25$	Q055 $x^2 + y^2 - 2x + 8y - 8 = 0$	Q081 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$
Q004 $(x - 3)^2 + (y - 4)^2 = 36$	Q030 $(x + 4)^2 + (y - 1)^2 = 58$	Q056 A	Q082 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$
Q005 $(x + 2)^2 + (y - 6)^2 = 12$	Q031 $(x - 2)^2 + (y + 1)^2 = 5$	Q057 B	Q083 $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$
Q006 $x^2 + y^2 = 25$	Q032 $x^2 + y^2 = 25$	Q058 D	Q084 C
Q007 A	Q033 C	Q059 D	Q085 D
Q008 A	Q034 A	Q060 D	Q086 C
Q009 D	Q035 D	Q061 $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$	Q087 C
Q010 B	Q036 A	Q062 A	Q088 B
Q011 B	Q037 B	Q063 $(x - 4)^2 + (y + 2)^2 = 4$	Q089 $(x - 1)^2 + (y + 1)^2 = 25$
Q012 A	Q038 A	Q064 $(x + 2)^2 + (y - 3)^2 = 4$	Q090 $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$
Q013 circle; centre (3, -1), radius 2	Q039 $(x - 2)^2 + (y + 1)^2 = 17$	Q065 $(x + 1)^2 + (y - 4)^2 = 16$	Q091 $(x - 2)^2 + (y - 3)^2 = 2$
Q014 circle; centre (2, 3), radius 3	Q040 $(x - 4)^2 + (y - 1)^2 = 10$	Q066 $(x - 3)^2 + (y + 4)^2 = 16$	Q092 $(x - 4)^2 + (y - 2)^2 = 26$
Q015 circle; centre (2, -4), radius 4	Q041 $(x - 1)^2 + (y - 3)^2 = 25$	Q067 $(x - 5)^2 + (y + 4)^2 = 16$	Q093 D
Q016 circle; centre (2, 0), radius 2	Q042 $(x + 1)^2 + (y + 2)^2 = 25$	Q068 $(x + 1)^2 + (y - 6)^2 = 1$	Q094 C
Q017 circle; centre (-1, 2), radius 5	Q043 $(x + 2)^2 + (y - 1)^2 = 2$	Q069 $(x - 2)^2 + (y + 5)^2 = 4$	Q095 C
Q018 circle; centre (-2, 3), radius $\sqrt{13}$	Q044 $x^2 + (y - 2)^2 = 25$	Q070 $(x + 3)^2 + (y + 4)^2 = 9$	Q096 D
Q019 A	Q045 A	Q071 B	Q097 $(x + 1)^2 + y^2 = 5$; centre (-1, 0), radius $\sqrt{5}$
Q020 C	Q046 C	Q072 B	Q098 C
Q021 D	Q047 C	Q073 B	Q099 $k < -1$ or $k > 3$
Q022 C	Q048 A	Q074 D	Q100 $k < -\sqrt{6}$ or $k > \sqrt{6}$
Q023 C	Q049 B	Q075 D	Q101 $k < 5/4$
Q024 C	Q050 C	Q076 D	Q102 $k < -1/5$ or $k > 3$
Q025 centre (5, 3), radius $^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$	Q051 $x^2 + y^2 - 5x - y + 4 = 0$	Q077 B	Q103 $-1 < k < 4$
Q026 B	Q052 $x^2 + y^2 - 3x - 7y + 6 = 0$	Q078 C	Q104 C

Q105 B	Q137 $-5 \leq m \leq 5$	Q169 $k = -1$ or $k = 7$	Q201 D
Q106 C	Q138 $r = \sqrt{2}$	Q170 A	Q202 B
Q107 B	Q139 $r > \sqrt{17}$	Q171 $4x + 3y = 25$	Q203 A
Q108 D	Q140 D	Q172 $5x + 3y = 34$	Q204 D
Q109 $0 < k < 4$	Q141 B	Q173 $3x - y = 10$	Q205 B
Q110 B	Q142 B	Q174 $-5x - 2y = 29$	Q206 C
Q111 $(1, -1)$ and $(-1, 3)$	Q143 B	Q175 $-2x + y = 5$	Q207 $(x - 3)^2 + (y - 1)^2 = 45$
Q112 $(0, 2)$ and $(2, 0)$	Q144 C	Q176 A	Q208 $(x + 3)^2 + (y + 9)^2 = 16$
Q113 $(3, -1)$	Q145 A	Q177 C	Q209 $(x - 3)^2 + (y + 9)^2 = 144$
Q114 $(-2, 1)$	Q146 C	Q178 A	Q210 A
Q115 $(-2, 1)$ and $(4, -1)$	Q147 B	Q179 C	Q211 D
Q116 C	Q148 D	Q180 A	Q212 A
Q117 D	Q149 $0 < r < 2$	Q181 $2x + y = 5$ and $-x + 2y = 5$	Q213 $(x + 1)^2 + (y + 2)^2 = 9$
Q118 C	Q150 D	Q182 $-3x + y = 10$ and $x + 3y = 10$	Q214 D
Q119 D	Q151 $\sqrt{14}$	Q183 $-2x + y = 5$ and $x + 2y = 5$	Q215 $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)
Q120 B	Q152 4	Q184 $y = 1$ and $4x - 3y = 5$	Q216 $x + y - 1 = 0$
Q121 0 points	Q153 $2\sqrt{5}$	Q185 $-x + 2y = 10$ and $2x + y = 10$	Q217 $12x - 6y - 7 = 0$
Q122 2 points : $(1, -3)$ and $(3, 1)$	Q154 $3\sqrt{2}$	Q186 D	Q218 $x = 1/2$
Q123 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$	Q155 2	Q187 A	Q219 A
Q124 0 points	Q156 A	Q188 D	Q220 C
Q125 1 point : $(-2, 0)$	Q157 C	Q189 B	Q221 B
Q126 D	Q158 D	Q190 C	Q222 D
Q127 A	Q159 B	Q191 $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 0)$	Q223 $(-3, -1)$ and $(-1, 3)$
Q128 C	Q160 D	Q192 B	Q224 $(0, 2)$ and $(8/5, -6/5)$
Q129 C	Q161 $k = \pm 2$	Q193 intersect at 2 points	Q225 $(1, 2)$ and $(2, 1)$
Q130 D	Q162 $k = \pm\sqrt{31}$	Q194 B : touch externally	Q226 B
Q131 $-5 < k < 5$	Q163 $k = \pm 2$	Q195 E : intersect at 2 points	Q227 C
Q132 $r = \sqrt{5}$	Q164 $k = 0$ or $k = -2$	Q196 C : touch internally	Q228 B
Q133 $m = \pm\sqrt{3}$	Q165 C	Q197 C : touch internally	Q229 $(-4, 3)$ and $(3, -4)$
Q134 $0 < r < \sqrt{10}$	Q166 D	Q198 E : intersect at 2 points	Q230 C
Q135 $m = \pm 4$	Q167 A	Q199 D : one is inside the other	Q231 $x^2 + y^2 + 2x + 6y - 15 = 0$
Q136 $n < -10$ or $n > 10$	Q168 B	Q200 C	Q232 $x^2 + y^2 + 4x + 2y = 0$

Q233 $x^2 + y^2 + 10x + 20y - 39 = 0$

Q238 D**Quiz MCQ 1** B**Quiz WRITTEN 6** $(4, 3), (-4, 3)$

Q234 $x^2 + y^2 + 24x + 32y - 5 = 0$

Q239 C**Quiz MCQ 2** D**Quiz WRITTEN 7** $3x + 4y = 25$

Q235 $x^2 + y^2 - 10x - 6y + 24 = 0$

Q240 B**Quiz MCQ 3** A**Quiz WRITTEN 8** $\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$ **Q236** D

Q241 $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Quiz MCQ 4 D**Q237** A**Q242** A**Quiz WRITTEN 5** $(x - 1)^2 + (y + 2)^2 = 25$